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Games, Virtual Worlds and Virtual Realities

5.1 Conceptualisation and History of VWs and Games

(a) Definitions and a Brief Historical Account

The concept of virtual worlds (VWs) pre-dates the Internet. The history of VWs began with text-based, offline role-playing games, created based on different works of fiction, such as, for instance, JRR Tolkien's books and the idea of world-building.¹ The first text-based interactive computer game appeared in 1970, *Colossal Cave Adventure*, with real-time interactive computer games called multi-user dungeons (MUDs) appearing by the end of the 1970s.² These were the first VWs. *MUD1* was created by Richard Bartle and Roy Trubshaw in 1979, at Essex University, being the first online network computer game. However, the most famous game in this group (text-based VWs) was *LambdaMOO*, created by Pavel Curtis in 1990.³

The literature analysing social, economic, technological and legal aspects of virtual worlds starts from the late 1990s, concerning the text-based VWs,⁴ and continues throughout the 2000s, discussing visually represented

¹ FG Lastowka and D Hunter, 'Virtual worlds: A primer' in JM Balkin and BS Noveck (eds), *The State of Play: Laws, Games, and Virtual Worlds* (NYU Press 2006) 13–28, 17–18; W Erlank, *Property in Virtual Worlds* (PhD thesis, Stellenbosch University 2012) 22–3 <<http://ssrn.com/abstract=2216481>> last accessed 1 August 2021.

² FG Lastowka and D Hunter, 'The laws of the virtual worlds' (2004) 92 *California Law Review* 1, 17.

³ F Rex, 'LambdaMOO: An introduction' *LambdaMOO* <<http://www.lambdamoo.info>> last accessed 1 August 2021. For more details about the history and the development of computer games in general, see for example J Juul, 'A history of the computer game' *Blog* (2001) <<http://www.jesperjuul.net/thesis/2-historyofthecomputergame.html>> last accessed 1 August 2021; J Dibble, *My Tiny Life: Crime and Passion in a Virtual World* (Henry Holt 1998).

⁴ See for example R Bartle, 'Hearts, clubs, diamonds, spades: Players who suit MUDs' (1996) 1(1) *The Journal of Virtual Environments* <<http://mud.co.uk/richard/hcde.htm>> last accessed 1 August 2021.

VWs and massively multiplayer online playing games (MMOPGs). The focus of the early literature was mainly on the technical, philosophical and governance issues of MUDs. More substantive legal discussion started at the beginning of the twenty-first century, with seminal works on the legal aspects of VWs, which tackle the following issues: economies and taxation;⁵ governance of VWs;⁶ property and IP;⁷ contracts and consumer

- ⁵ See for example E Castronova, 'On virtual economies' (2003) 3 *The International Journal of Computer Gaming Research* <<http://www.gamestudies.org/0302/castronova/>> last accessed 1 August 2021; E Castronova, 'Real products in imaginary worlds' (2005) 83(5) *Harvard Business Review* 20, 20–2; E Castronova, 'The right to play' (2004) 49 *New York Law School Law Review* 185; E Castronova, 'Virtual world economy: It's Namibia, basically' *Terranova* (3 August 2004) <http://www.terranova.blogs.com/terra_nova/2004/08/virtual_world_e.html> last accessed 1 August 2021; Lastowka and Hunter, 'Virtual worlds' (n 1) 9; MA Cherry, 'A taxonomy of virtual work' (2011) 45 *Georgia Law Review* 951.
- ⁶ See for example Lastowka and Hunter, 'Virtual worlds' (n 1); V Mayer-Schoenberger and JR Crowley, 'Napster's second life? – The regulatory challenges of virtual worlds' (2006) 100 *Northwestern University Law Review* 1775; BJ Gilbert, 'Getting to conscionable: Negotiating virtual worlds' end user license agreements without getting externally regulated' (2009) 4 *Journal of International Commercial Law and Technology* 238; JM Balkin, 'Virtual liberty: Freedom to design and freedom to play in virtual worlds' (2004) 90(8) *Virginia Law Review* 2043.
- ⁷ Lastowka and Hunter, 'Virtual worlds' (n 1); SH Abramovitch and DL Cummings, 'Virtual property, real law: The regulation of property in video games' (2007) 6(2) *Canadian Journal of Law and Technology* 73; J Fairfield, 'Virtual property' (2005) 85 *Boston University Law Review* 1047; E Castronova, 'Virtual worlds: A first-hand account of market and society on the cyberian frontier' (2001) 618 CESifo Working Papers Series; C Blazer, 'The five indicia of virtual property' (2006) 5 *Pierce Law Review* 137; R Vacca, 'Viewing virtual property ownership through the lens of innovation' (2008) 76 *Tennessee Law Review* 33; TJ Westbrook, 'Owned: Finding a place for virtual world property rights' (2006) *Michigan State Law Review* 779–812; Erlank (n 1); JM Moringiello, 'Towards a system of estates in virtual property' (2008) 1 *International Journal of Private Law* 3; *Kremen v Cohen*, 337 F.3d 1024 (9th Cir 2003); A Chein, 'A practical look at virtual property' (2006) 80 *St. John's Law Review* 1059; C Cifrino, 'Virtual property, virtual rights: Why contract law, not property law, must be the governing paradigm in the law of virtual worlds' (2014) 55 *Boston College Law Review* 235; MG Veloso III, 'Virtual property rights: A modified usufruct of intangibles' (2010) 4 *Philippine Law Journal* 82; J Slaughter, 'Virtual worlds: Between contract and property' (2008) 62 *Yale Student Scholarship Papers* <http://digitalcommons.law.yale.edu/student_papers/62> last accessed 1 August 2021; U Yoon, 'South Korea and indirect reliance on IP law: Real money trading in MMORPG items' (2008) 3(3) *Journal of Intellectual Property Law and Practice* 174; JW Nelson, 'Fiber optic foxes: Virtual objects and virtual worlds through the lens of *Pierson v Post* and the law of capture' (2009) 14 *Journal of Technology Law & Policy* 5; JW Nelson, 'The virtual property problem: What property rights in virtual resources might look like, how they might work, and why they are a bad idea' (2010) 41 *McGeorge Law Review* 281.

protection;⁸ and virtual crime.⁹

There is a variety of academic literature discussing death and VWs from anthropological, sociological, psychological, educational and other perspectives,¹⁰ but little, almost nothing, from a legal angle. Literature about VWs and games has been decreasing steadily in the last two decades. However, issues around their legal nature have remained.

Legal aspects of transmission of other digital assets on death (e.g. emails, social network accounts, online banking accounts, photos, domain names) have been explored to an extent following the growing importance of these assets in life and on the death of the users, only sporadically mentioning VW accounts as types of digital assets.¹¹ This literature has been explored in more detail in other chapters.

This chapter addresses the gap in the literature and sheds light on the post-mortem legal status of different in-game assets (e.g. avatars, weapons, houses, land). I have addressed this issue in my previous writings, and this

⁸ A Jankowich, 'The complex web of corporate rule-making in virtual worlds' (2006) 8 *Tulane Journal of Technology and Intellectual Property* 1, 52–3; B Glushko, 'Tales of the (virtual) city: Governing property disputes in virtual worlds' (2007) 22 *Berkeley Technology Law Journal* 507; R Shikowitz, 'License to kill: MDY v Blizzard and the battle over copyright in World of Warcraft' (2009–10) 75 *Brooklyn Law Review* 1015; J Fairfield, 'Anti-social contracts: The contractual governance of virtual worlds' (2007) 53 *McGill Law Journal* 427; D Miller, 'Determining ownership in virtual worlds: Copyright and license agreements' (2003) 22 *Review of Litigation* 435; P Riley, 'Litigating Second Life land disputes: A consumer protection approach' (2009) 19(3) *Fordham Intellectual Property, Media and Entertainment Law Journal* 877; PJ Quinn, 'A click too far: The difficulty in using adhesive American law license agreements to govern global virtual worlds' (2010) 27 *Wisconsin International Law Journal* 757.

⁹ See for example OS Kerr, 'Criminal law in virtual worlds' 2008 *University of Chicago Legal Forum* 415; Lastowka and Hunter, 'Virtual worlds' (n 1); AR Lodder, 'Dutch Supreme Court 2012: Virtual theft ruling a one-off or first in a series?' (2013) 6(3) *Journal of Virtual Worlds Research* 1; AV Arias, 'Life, liberty and the pursuit of swords and armor: Regulating the theft of virtual goods' (2008) 57 *Emory Law Journal* 1301.

¹⁰ See for example R Ferguson, 'Death of an avatar: Implications of presence for learners and educators in virtual worlds' (2012) 4(2) *Journal of Gaming & Virtual Worlds* 137; M Jakobsson, 'Rest in peace, Bill the bot: Death and life in virtual worlds' in R Schroeder (ed.), *The Social Life of Avatars* (Springer 2002); JA Archinaco, 'Virtual worlds, real damages: The odd case of American Hero, the greatest horse that may have lived' (2007) 11(1) *Gaming Law Journal*; CJ Sofka, IN Cupit and KR Gilbert (eds), *Dying, Death, and Grief in an Online Universe: For Counselors and Educators* (Springer 2012); A Haverinen, *Memoria Virtualis – Death and Mourning Rituals in Online Environments* (PhD thesis, University of Turku 2014) <<https://www.doria.fi/handle/10024/98454>> last accessed 1 August 2021.

¹¹ The literature includes my own and other work cited throughout this book.

analysis will be primarily based on the arguments and analysis set out therein.¹² However, this analysis is more thorough, and the chapter addresses the topic at a greater length and depth than any of the earlier articles.

Following this book's principal and subordinate research questions, the chapter explores how the notion of property applies to VWs and games. It will attempt to determine if there is property in VWs and games, and if not, what would be an appropriate legal treatment of user rights and interests in virtual, in-game assets. The aim is to determine the possibility of transmission of virtual assets (VAs) or their value on the death of a player.

The term 'virtual assets' (VAs) is used here provisionally, trying to avoid any implications about the potential legal nature of these assets. Therefore, the analysis avoids the widely used term 'virtual property' (VP) at the moment. Later in this chapter, after exploring different legal concepts that could characterise these assets, the term might change and adapt to the findings. Until then, the term 'VAs' will be used to describe any item, object or asset found in VWs or games and used or created by players (e.g. avatars, weapons, land, houses, clothes, furniture, and anything else that could be found in different VWs). It is also vital to differentiate VAs from digital assets defined in the Introduction. VAs are only a subset of digital assets (DAs), defined as any asset of value online, capable of post-mortem transmission.

Therefore, the analysis will assess whether these assets could fit within the notions of property or some other relevant legal concepts (IP, servitudes, easements, the right of use), which would result in their being recognised as part of a user's estate. This first question aligns with the book's theoretical underpinning, that is, the discussion of whether a digital asset can and should be considered property and, consequently, transmit on death. The discussion and conceptual framework of PMP are not included in this chapter because currently, sharing and storing personal data is not a predominant feature of VWs, as it is in emails and social networks. Avatars or other items represent VWs and online game users, who usually do not publicly share their identities and personal data.

I do not share the views of most of the classical legal VWs literature, arguing for the recognition of 'virtual property' and referring to full ownership. Instead, this chapter proposes a compromise solution to reconcile different interests arising in VWs, primarily those of developers and players. Recognising a phenomenon of *constitutionalisation* of VWs and arguing for

¹² E Harbinja, 'Virtual worlds players – consumers or citizens?' (2014) 3(4) *Internet Policy Review* <<http://policyreview.info/articles/analysis/virtual-worlds-players-consumers-or-citizens>> last accessed 1 August 2021; E Harbinja, 'Virtual worlds – a legal post-mortem account' (2014) 11:3 *SCRIPTed* 273.

better recognition of players' in-game interests, the chapter identifies a solution entitled 'VWs usufruct', a personal right of a player against the VW provider in second layer assets. If VA exchanges are legal on recognised auction sites, the right will only be transmissible as a monetary claim. If no such auction sites exist or ToS do not permit them, then monetisation is impossible, and neither is the right to compensation. The solution would enable players to control their virtual assets and heirs to benefit from potentially valuable VW accounts and VAs.

It is also important to note that findings in this chapter will apply not only to VWs, defined more narrowly, but also to online games, widely played across the world and controlled by service providers. We will not consider video games only played locally at the user end, which excludes more meaningful interactivity online and a degree of permanence and stability.

(b) Conceptualisation of VWs and Games

From a linguistic perspective, VWs could be defined as states of human existence, which do not exist physically, are not real, but appear to be real from the program's point of view or that of the user.¹³ From this definition, we could extract the most critical features that define VWs: computer-moderated; persistence; environmental attributes (immersive and persuasive worlds, mimicking the real world); interactivity; and the participation of multiple individuals.¹⁴

Developers use different business models for their VWs. Some of them are closed, used for military or business simulations, whereas others are open, commercial worlds, where users can join for free, for monthly fee payment (e.g. *World of Warcraft*), or operate on a freemium basis (e.g. *Second Life*), where the basic services are free and others have their price.¹⁵

The umbrella term for VWs is MMOPGs, but these can be divided into social and game worlds based on their player community and structure. In game VWs (massively multiplayer online role-playing games – MMORPGs), players take a specific role and compete to achieve certain predefined goals (e.g. *World of Warcraft*). In the social or unstructured worlds, the emphasis is on social interaction with other players and the environment (e.g. *Second Life*, *IMVU*). Therefore, these VWs are not games but platforms for social

¹³ *Oxford English Dictionary (OED)* (online edn, Oxford University Press 2016) last accessed 1 August 2021.

¹⁴ Erlank (n 1) 47–57.

¹⁵ See J Fairfield, 'The end of the (virtual) world' (2009) 112(1) *West Virginia Law Review* 53; Riley (n 8).

interaction or 'mirror worlds'.¹⁶ The third kind of VWs is kids' worlds, those targeting children as the main player base (e.g. Club Penguin).¹⁷

According to the technology employed to enable access to the worlds, the worlds are divided into client-based (e.g. *World of Warcraft*) and those where players can join simply online (e.g. *Second Life*). Some video games, including some VWs (e.g. *The Lord of the Rings Online*, *Dungeons & Dragons Online*, *EverQuest II*, *Diablo*), can also be accessed from intermediaries, the most prominent of which is an entertainment platform called Steam.¹⁸

This chapter will focus on two examples: *World of Warcraft* and *Second Life*. The reason for choosing the US-based VWs is that most of the successful Western VWs are indeed hosted in the US,¹⁹ choice of law provisions usually points to US law, and most common law cases have been resolved there.²⁰ Also, these examples are chosen for their historical domination in the market, previous and retained user base, their impact and their 'cultural footprint'.²¹ *Second Life* has been steadily declining in popularity, but it is still worth mentioning as most of the existing case law involves this VW and its historical and cultural significance and replicability to similar gaming models in the future.²²

Before initiating the virtual property and contracts analysis, and in order to bring potential legal disputes closer to the reader, the following section will present some examples. These examples are actual US court cases, and they serve to illustrate legal issues appearing in VWs. Unfortunately, at the time of writing, there has not been a single UK case discussing the issues of VWs and VP.

¹⁶ KZero Worldwide, 'Radar charts Q2 2014 VWs and MMOs shown by genre, average user age and status' *KZero* (2004) <<http://www.kzero.co.uk/blog/category/education-and-academia>> last accessed 1 August 2021.

¹⁷ G Lastowka, *Virtual Justice: The New Laws of Online Worlds* (Yale University Press 2010) 58.

¹⁸ The platform distributes different video games and other software, from both independent and established software companies. It is also a communication, social networking and multiplayer platform, allowing different kinds of interactions between players (akin to social network sites). The further evolution of VWs includes innovative hardware (e.g. Oculus Rift), bringing even more reality to these worlds. See KZero Worldwide, 'Consumer virtual reality: State of the market report' *KZero* (2014) <<http://www.kzero.co.uk/blog/category/education-and-academia>> last accessed 1 August 2021.

¹⁹ B Edwards, 'The 11 most influential online worlds of all time' *PCWorld* (2011) <<http://www.pcworld.com/article/228000/influentialonlineworlds.html>> last accessed 1 August 2021.

²⁰ Fairfield, 'Anti-social contracts' (n 8) 430.

²¹ Quinn (n 8).

²² Sporadic references will be made to other VWs and platforms, but the main analysis will be based on the examples of these two VWs.

5.2 Illustrations through Case Law

(a) *Bragg Case: Property and Jurisdiction in VWs*

The first and most famous VWs case is that of *Bragg v Linden Research, Inc.*²³ In this case, Marc Bragg sued the owner of *Second Life*, Linden Research, after it expelled him from the online community and reclaimed his VAs, ‘effectively confiscating all of the virtual property and currency that he maintained on his account’ (roughly \$2,000 in real-world money on account).²⁴ Linden Lab expelled Marc Bragg, claiming that he had violated its terms of service by improperly buying land at an auction for approximately \$300. *Second Life* moved to compel arbitration according to the terms of service agreement. Bragg, however, argued that the contractual terms between Bragg and *Second Life* were unconscionable because the service agreement assumed too much power and was unreasonably biased against the user. The court confirmed that the terms of service were unconscionable in relation to the mandatory arbitration clause and knocked it down.²⁵ More specifically, the court focused on the fact that there was a surprise due to hidden or missing terms because there was no notice of the serious expense and inconvenience to the plaintiff, having to spend ten to twenty thousand dollars to pay the arbitrators in addition to having to go to California from Pennsylvania in order to take part in the arbitration. The court said that the terms left the plaintiff with no effective remedy.²⁶ The court applied the Californian law in its analysis of the contract and noted that it must find both procedural and substantive unconscionability in California to find unconscionability.²⁷ It found both elements and concluded that the arbitration clause was unconscionable.²⁸

This case has not decided on the issue of VP. Bragg initially brought up the property claim, asserting that his in-game assets are his property (his land,

²³ 487 F.Supp.2d 593, 612 (ED Pa 2007).

²⁴ Ibid. 611.

²⁵ Unconscionable terms are those ‘so extreme as to appear unconscionable according to the mores and business practices’ used at the time. See N Kutler, ‘Protecting your online you: A new approach to handling your online persona after death’ (2011) 26 *Berkeley Technology Law Journal* 1641; UCC § 302-2 (amended 2003); *Williams v Walker-Thomas Furniture Co.*, 350 F.2d 445, 449 (DC Cir 1965); *Carnival Cruise Lines, Inc v Shute*, 499 US 585, 600–1 (1991).

²⁶ *Bragg v Linden Research, Inc* (n 23) 611. For more, see S Hetcher, ‘User-generated content and the future of copyright: Part two – agreements between users and mega-sites’ (2008) 24 *Santa Clara Computer & High Technology Law Journal* 829, 836.

²⁷ *Bragg v Linden Research, Inc* (n 23) 605, 606.

²⁸ Ibid. 611.

fireworks, and other items he possessed).²⁹ The court, unfortunately, did not discuss it, so VP, as demonstrated later, remains on the level of academic debates.

(b) *Evans Case: Account Suspension and Settlement*

The second case, involving *Second Life* and Linden, lost another chance to discuss VP by a court and provide us with some guidance on this issue. In *Evans et al. v Linden Research, Inc et al.*,³⁰ the central issue was fairness and validity of the contract (provisions about the suspension of accounts and users compensation). A group of users claimed to own their VAs³¹ and complained that they had purchased virtual items and/or virtual land and later had their accounts unilaterally terminated or suspended by Linden. These players were not compensated for the value of the virtual land, items and/or currency in their accounts. In addition, the plaintiffs claimed that Linden made false representations about ownership of virtual land and virtual items and wrongfully confiscated these items from the class members they sought to represent.³² Linden disputed the claimed ownership in VAs, recognising IP rights in users' creations, copyright in particular. It maintained that copyright in users' creations is indisputable; however, it has been licensed to Linden, and it is entitled to remove a licensed copy from its servers.³³

Again, there was no decision in respect of VP. The case was settled,³⁴ and Linden agreed to do the following: return up to 100 per cent of the US dollar balances to the PayPal accounts of the plaintiffs; return up to 100 per cent of the Linden dollar balances in class members' accounts; pay two Linden dollars per square metre of virtual land held by class members; and regarding virtual items, to pay \$15 per class member to their PayPal account or let the class members attempt to sell their virtual items on the *Second Life* Marketplace, at which point Linden will waive *Second Life's* commission on the sales.³⁵ This example might illustrate Linden's attitude and concerns over VP and willingness to compensate the users instead of proceeding with the case, which might find some property in virtual items and land and have unwanted consequences for Linden (user's control, transfer, transmission and other incidents

²⁹ Ibid. 585.

³⁰ No C-11-01078 DMR (United States District Court, ND Cal 20 November 2012).

³¹ Hearing Transcript 27:12–28:11.

³² Ibid.

³³ Ibid. 37:7–10; 39:17–24; 53:15–24.

³⁴ *Evans et al., Plaintiffs, v Linden Research, Inc et al.*, No C-11-01078 DMR (United States District Court, ND Cal 25 October 2013).

³⁵ See the confirmation of settlement in *Evans et al. v Linden Research, Inc et al.* No C-11-01078 DMR (United States District Court, ND Cal 20 November 2012).

identified in Chapter 2). On the other hand, the fact that the court, as in the Bragg case, avoided a decision on VP and turned to contracts and IP, demonstrates again their reluctance to engage and make such a radical legal and policy decision. This reasoning is justifiable from a normative and doctrinal perspective, as suggested later in this chapter.

In addition to the Bragg and Evans cases, there have been two interesting virtual theft³⁶ cases in the Netherlands³⁷ and one in China.³⁸ However, these cases are examples specific to these civilian countries and should not be regarded as an indication of a general approach. In these jurisdictions' absence of relevant case law, it is uncertain whether the US or English criminal courts would recognise VP in theft cases. Moreover, even if, eventually, cases were prosecuted in the courts of the jurisdictions relevant to this book, the outcomes would not provide a firm argument for VP. This is because the rationale and definition of property in criminal cases are quite different from civil cases. Consequently, there is limited value in considering these cases and criminal law herein. In the lack of case law worldwide, the cases have been mentioned here to illustrate different outlooks on VP, rejecting their substantive relevance for this book.

To summarise, the scarce Western VWs case law illustrates the potential disputes that might arise in VWs, namely ownership, property claims and disputes over VW contract provisions. However, the cases have only opened up the floor to more debates and do not determine whether there is property in VWs. Furthermore, the disparate approaches in these cases are based on very different legal and cultural traditions. Finally, the cases focused on the second layer (see the classification in the section below, i.e. different items in *Second Life*, such as fireworks, avatars, weapons or land) and this layer, as demonstrated later in this chapter, is the most important, problematic and worth examining in detail. Also, it is interesting that Linden explicitly

³⁶ See more in JW Nelson, 'A virtual property solution: How privacy law can protect the citizens of virtual worlds' (2011) 36 *Oklahoma City University Law Review* 395, 413; Arias (n 9) 1306–7.

³⁷ Kerr (n 9) 422–3; BBC News, 'Virtual theft leads to arrest' (14 November 2007) <<http://news.bbc.co.uk/1/hi/7094764.stm>> last accessed 1 August 2021; Lodder (n 9) 3, 4, 8.

³⁸ W Knight, 'Gamer wins back virtual booty in court battle' *New Scientist* (23 December 2003) <<http://www.newscientist.com/article/dn4510-gamer-wins-back-virtual-booty-in-courtbattle.Html>> last accessed 1 August 2021; J Lyman, 'Gamer wins lawsuit in Chinese court over stolen virtual winnings' *TECHNEWSWORLD* (19 December 2003) <<http://www.technewsworld.comystory/3244L.html>> last accessed 1 August 2021; Shikowitz (n 8) 51; Glushko (n 8) 518; *Li v Beijing Arctic Ice Tech Dev Co* (Beijing Chaoyang Dist People's Ct 18 December 2003) <<http://www.chinacourt.org/public/detail.php?id=143455>> last accessed 1 August 2021.

admitted players' IP rights in the Evans case (the third layer, according to the classification below). Therefore, this layer is much less disputable and will not be discussed in detail.

Notwithstanding the vagueness left by these cases, this chapter will assess whether there is property in VWs, who owns this property and, finally, if existent, how this property transmits on death. Before discussing these issues, it is crucial to identify the potential objects of VP, that is, virtual assets. The following section will present a classification of VAs as possible objects of property in VWs and games more generally. This classification will be used as a basis for the analysis throughout this chapter.

5.3 Layers of Virtual Assets

Most virtual property theories tend to confuse different types of code and content in VWs, equating the underlying software (the building blocks of VWs) and the user-generated content (virtual assets). In this regard, Abramovitch offers a helpful theory and proposes three layers (or levels) where property or VAs can be identified within VWs.³⁹

At the *first layer*, there is the developer's code, protected by IP as software. This level, therefore, represents software and code that determines properties, features of VWs, and users' actions and behaviours.

At the *second layer*, Abramovitch identifies objects or items inside the VW which resemble real-world items (objects like avatars, weapons, buildings, clothing, cars, spaceships, houses). This layer mimics real-world objects and includes a perception of physicality from a user's point of view.

At the *third layer*, she identifies in-game VAs that could be protected by IP (e.g. a book found lying on a table inside the VW, paintings, statues). This is essentially the user's creative work inside a VW, and it is different from the second layer in that it may lack physicality and does not mimic real-world objects. The difference between the two layers can be further analogised with the division of property rights in a physical object and copyright in a work embedded in that physical object.

The layer approach is helpful for the analysis in this book for two main reasons: first, it is more nuanced as an approach and does not represent a unified, rigid 'player-deserves-all' (VP should belong to the players) or 'the developer-deserves-all' perspective (property in servers or IP in software should extend to the virtual realm), usually found in the early 2000s literature. These two approaches fail to recognise, on the one hand, *constitutionalisation* of virtual

³⁹ SH Abramovitch, 'Virtual property in virtual worlds' *Gowlings.com* (2009) 1–2 <<http://www.gowlings.com/knowledgecentre/publicationPDFs/TLI-2009-Susan-Abramovitch-Virtual-Property-in-Virtual-Worlds.pdf>> last accessed 1 August 2021.

worlds and their significance for the player; and, on the other, the IP interests of the developer.

Second, this approach acknowledges the Internet architecture and the fact that significant investments are made by world owners, arguing simultaneously for assessing the rights of users at a different game level.

Third, this differentiation offers a possibility to discuss and suggest recognising various legal concepts at different code and virtual reality levels and offer some compromise and more widely acceptable legal solutions.⁴⁰ Protection for the different layers can be provided for by IP, property, limited real rights (rights on the property of another) or contracts, depending on the characteristics of an individual layer.⁴¹

The analysis in this chapter will accept and use this classification, focusing primarily on the second layer.⁴² This layer has been discussed in the court cases illustrated earlier. Of course, the courts have not identified VAs using this classification, but the disputes in the Bragg and Evans cases concerned VAs such as land and other different items belonging to our second level.

The reason for excluding in-depth discussion of the first layer is that it is much less disputed, and the underlying code indeed belongs to the developer (protected by copyright or patent in software and property in physical servers).⁴³ The first layer will be discussed only to the extent it relates to or determines the second and third levels. Apart from having a more straightforward legal nature, the first level is beyond the scope of this chapter, as the book looks at the player's or user's ability to transmit their virtual assets on death. The first layer assets are a company's assets, and their succession is regulated differently, using company law rules.

The third layer will be mentioned sporadically, but IP issues will not be analysed in detail. This book considers novel issues in the transmission of

⁴⁰ Erlank (n 1) 182.

⁴¹ Ibid.

⁴² For more details about the copyright protection in VWs, see S Roncallo-Dow et al., 'Authorship in virtual worlds: Author's death to rights revival?' (2013) 6(3) *Journal of Virtual Worlds Research* 1; Miller (n 8).

⁴³ See for example cases such as *SAS Institute v World Programming* C-406/10 and *Nova Productions v Mazooma Games* [2007] RPC 25, which suggest that graphics in computer games could be regarded as artistic works and protected by copyright. In the US, the courts have developed the doctrine of 'cybertrespass' to companies' computer systems or servers, extending the tort of trespass to cyberworld. See *eBay v Bidder's Edge Inc*, 100 F.Supp.2d 1058 (ND Cal 2000); *Ticketmaster Corp. et al. v Tickets.com, Inc*, No CV 99-7654 (CD Cal 27 March 2000). For more, see M Carrier and G Lastowka, 'Against cyberproperty' (2007) 22 *Berkeley Technology Law Journal* 1485.

digital assets on death; transmission of IP rights to heirs is not in dispute if players' creations meet copyright requirements.

In order to emphasise the main distinction between VWs and games and other assets discussed in this book, property and proprietary rights in the second layer will be the focus. The main reason for this is the peculiar aspects of the assets, namely, physicality.

5.4 Virtual Property

Virtual property is a theoretical construct about property rights in the items and resources originating and existing in VWs and games. Much has been written *pro* and *contra* the recognition of virtual property. However, it still exists primarily in academic discussions, and courts or legislators have not recognised its importance. There have been some judicial attempts to address virtual property (see the Bragg and Evans cases, above), but there have not been any legislative efforts at all. This section aims to shed some more light on VP, exploring whether there should be property rights in VWs, and if not, what the potential alternatives are.

(a) *The Legal Status of Virtual Property*

This section will look at whether virtual property and virtual assets share the features of property and property objects identified in Chapter 2.

First, the discussion will assess whether Honoré's and Becker's incidents of property could be found in VP. Second, we will explore the leading analysis of VP and its features, that is, Fairfield's theory. He lists three major criteria: incidents of property, borrowing from the law, and economics literature. Coinciding with those identified in the second chapter, the main incidents are rivalrousness, permanence and interconnectedness. Castronova et al. use the same incidents as those inherent in physical objects, attempting to define and justify VP.⁴⁴ Some identify further incidents (such as scarcity,⁴⁵ secondary markets and 'value-added-by-users'⁴⁶). The analysis in the following section will add tangibility to this list as an essential feature of property historically and still retained as such by some jurisdictions (e.g. England).⁴⁷

⁴⁴ 'Virtual worlds are virtual because they are online, but they are worlds because there is some physicality to them.' See Castronova, 'On virtual economies' (n 5).

⁴⁵ Ibid. Some authors, however, claim that scarcity is artificially created and coded, and usually for the reasons of the provider's economic benefits. See Lastowka (n 17) 135; Erlank (n 1) 270, 271.

⁴⁶ Blazer (n 7) 139.

⁴⁷ Harbinja, 'Virtual worlds – a legal post-mortem account' (n 12) 273.

Honoré's Incidents

As discussed in Chapter 2, one of the most influential theories of property incidents has been offered by Honoré. His eleven exhaustive incidents (the right to possess, the right to use, the right to manage, the right to the income of the thing, the right to the capital, the right to security, the rights of transmissibility and absence of term, the prohibition of harmful use, the liability of execution, and the incident of residuary) could be applied to our VP layers. For the first layer (owned by developers), all of the incidents are present.

Looking at the second level VP, only a few of these seem to be present, namely the right to possess, the right to use, the prohibition of harmful use, and the right to the income of the thing (to an extent, depending on the type of the VW). Most of the other incidents are in the hands of the providers (i.e. the right to the capital, the right to security, the rights of transmissibility and absence of term, the liability of execution, the incident of residuary, and the right to manage). Ownership of the third level VP encounters similar difficulties, and this layer possesses the same incidents as the second one. In addition, it could be argued that it possesses the right to transmissibility (as IP does), but not the absence of term, since most contracts recognise IP in players' creations without recognising property at the same time. If we were to recognise VP, these incidents, according to Honoré, would need to be included for full ownership to exist.

Becker's Incidents

Building on Honoré's analysis, Becker proposes thirteen elements: the right (claim) to possess, the right (liberty) to use, the right (power) to manage, the right (claim) to the income, the right (liberty) to consume and destroy, the right (liberty) to modify, the right (power) to alienate, the right (power) to transmit, the right (claim) to security, the absence of term, the prohibition of harmful use, liability to execution, and residuary rules.⁴⁸ He further argues that anyone who has one of the first eight elements plus the right to security has a property right.

Again, there is no dispute about the first layer VAs. The second layer would lack the right (liberty) to consume and destroy (destroy in particular), the right (power) to alienate (for some VWs), the right (power) to transmit, the right (claim) to security, the absence of term, liability to execution, and residuary rules. Therefore, the bundle of eight rights is not present here, and VP would mean recognising these, according to this theory. Similarly, as

⁴⁸ L. Becker, 'The moral basis of property rights' in JR Pennock and JW Chapman (eds), *Property* (New York University Press 1980) 190–1.

discussed above, third layer VAs would lack all of these, apart from the right to transmit. Consequently, the theory cannot justify the creation of VP.

Incidents of Property Objects vs Virtual Property Objects

A potential problem VP in the second and third layers would encounter is its alleged lack of tangibility. This problem would not be as significant for the civil law countries, as they recognise property in intangibles, either in the civil code, like France,⁴⁹ or by establishing a separate category of property, constitutional property, like Germany.⁵⁰ This could be more difficult for the English common law system, which refuses to consider intangibles property, at least in some cases (information), but decides to recognise it in others, for example IP. US law is more likely to recognise intangibles as property (see the relevant cases on propertisation of information in Chapter 6).⁵¹

Therefore, the intangibility of second layer VAs (intangible for the purpose of the classical legal definitions, i.e. lack of real-world tangibility and incorporeality, as it consists of code) would present an obstacle for recognising VP in English common law. Fewer issues would emerge in the US, and civil law systems would require legislative interventions. To develop this point further, it might be suggested that this layer does not have to be considered intangible, as it seems tangible to an avatar in a game, and if the level of immersion of a player is very high, then consequently, it could be tangible for the player as well.⁵² At the moment, however, this line of thought is unlikely to be accepted by the English courts as a suitable legal test for tangibility, as demonstrated in Chapter 2.

Penner's analysis of property objects includes a requirement of 'separability' or 'thinghood', meaning that things, in order to be property, must not be conceived of as 'an aspect of ourselves or our on-going personality-rich

⁴⁹ J Bell et al. *Principles of French Law* (2nd edn, Oxford University Press 2008) 27.

⁵⁰ MJ Raff, *Private Property and Environmental Responsibility: A Comparative Study of German Real Property Law* (Kluwer Law International 2003).

⁵¹ See important cases about the propertisation of information: *Exchange Telegraph Co Ltd v Gregory & Co* [1896] 1 QB 147; *Oxford v Moss* [1978] 68 Cr App R 181. Generally, see N Palmer and P Kohler, 'Information as property' in N Palmer and E McKendrick (eds), *Interests in Goods* (Lloyd's of London Press 1993) 187–206. The US common and statute law is readier to recognise property in different types of intangibles (e.g. fresh news). The doctrine of misappropriation is of a proprietary character, as established in *International News Service v Associated Press*, 248 US 215, 250 (1918). See commentary in RY Fujichaku, 'The misappropriation doctrine in cyberspace: Protecting the commercial value of "hot news" information' (1998) 20 *University of Hawai'i Law Review* 421.

⁵² Erlank (n 1) 287–8.

relationships to others'.⁵³ This theory, for instance, would not consider personal data, body parts or blood as property, since they all lack the separability requirement and are aspects of ourselves. Applying this argument to our second layer, and immersion notwithstanding, it could be argued that avatars would lack these criteria whereas the rest of the VAs in this layer would not, as weapons, spaceships and clothes, for example, are separable both from the avatar and the player.

Similarly, the definition of property offered by Waldron focuses primarily on 'material resources', 'capable of satisfying some human need or want'.⁵⁴ This definition requires materiality and not tangibility, and it could be argued that second layer VAs are material resources within VWs, and they satisfy the player's need for entertainment, creation, competition, trade, education, and so on. However, the third layer VAs are less likely to be considered material resources, and this layer encounters the same difficulties as IP.

The analyses will further be based on the features identified by Fairfield in his seminal work on virtual property. The first feature he analyses is rivalrousness, which means that consumption cannot be shared for a rivalrous resource; one person's possession and consumption physically excludes the other pretenders to the same resource.⁵⁵

Fairfield thus discusses the possibility of applying the traditional concept of property, designed for chattels rather than IP, to virtual property that mimics the real, offline one (layer two VAs). He distinguishes between the computer, software code, designed as non-rivalrous (protected by IP, layer 1) and other types of code, rivalrous, 'designed to act more like land or chattel than ideas'⁵⁶ where if one person controls it, the others cannot.

Rivalrousness is, therefore, the physical quality of an object, different from exclusivity, which is an individual's power to control the use of an object.⁵⁷ Other commentators indeed use the term 'exclusivity' as a synonym for rivalrousness.⁵⁸ This is wrong, however, and Fairfield rightly notes that exclusivity is a function of rivalrousness, the quality that can be assigned to non-rivalrous objects by law or technology, for instance (e.g. IP creations and digital rights management). It is important to note that this code is rivalrous

⁵³ Ibid. 126.

⁵⁴ J Waldron, *The Right to Private Property* (Clarendon Press 1990).

⁵⁵ C Hess and E Ostrom, *Understanding Knowledge as a Commons from Theory to Practice* (MIT Press 2007) 352.

⁵⁶ Ibid. 1101.

⁵⁷ DL Weimer and AR Vining, *Policy Analysis* (5th edn, Longman 2011) 72.

⁵⁸ Westbrook (n 7).

because it is made that way and is a fundamental part of the Internet.⁵⁹ Examples of this code are domain names, URLs, websites, email accounts and VW items. Fairfield also warns of the confusion that occurs in trying to fit all intangibles into a category of non-rivalrous objects.⁶⁰ Other authors who support his stance concerning VP and rivalrousness are Horowitz,⁶¹ Blazer⁶² and Westbrook.⁶³

Critics claim that VP and VAs are non-rivalrous. Nelson, for instance, disputes rivalrousness, or rather, exclusivity, of virtual goods using the same examples as Fairfield, that is, URLs and emails. He claims that the alleged owner cannot control this property to the exclusion of others because, according to the contract that a user concludes, the developer retains the ability to control these resources. Similarly, Glushko argues that the ease of copying code in the case of any digital property would undermine the exclusivity of virtual property as well.⁶⁴ These authors have again confused the notions of exclusivity, which is an economic and legal feature and relates to the rights conferred by contracts or property, and rivalrousness, which is a purely physical feature, so even if a provider retains the exclusive control over a virtual resource, the fact that only one user at a time can, arguably, physically experience it means that the resource is rivalrous.

In summary, rivalrousness is a feature of the second level virtual property. The problem with this feature incident is its unstable nature, as it only exists if created in that form by the developer. However, this should not be an issue since, ultimately, VWs are unstable too and would not exist if developers did not create them. Indeed, players and many theorists (e.g. Lastowka or Mayer-Schönberger), myself included, still accept VWs as such, claiming that however unstable and peculiar places they may be, VWs still represent replicas of the real world. In addition, even if we accept that VW items are not rivalrous, this is not a decisive point to discard their protection since IP resources are not rivalrous either and still are protected like, or similar to, property.

Permanence or persistence of VWs and in-game assets is another disputed feature, present in physical property and contested in the case of IP. Castronova defines persistence as the feature of VWs enabling them to

⁵⁹ Fairfield, 'Virtual property' (n 7) 1053–4.

⁶⁰ Ibid. 1063.

⁶¹ SJ Horowitz, 'Competing Lockean claims to virtual property' (2007) 20(2) *Harvard Journal of Law and Technology* 443.

⁶² Blazer (n 7).

⁶³ Westbrook (n 7).

⁶⁴ Glushko (n 8) 251–7.

‘continue to run whether anyone is using [them] or not’.⁶⁵ Fairfield, like Castronova, argues that code is persistent too since ‘it does not fade away after each use, and it does not run on one single computer’.⁶⁶ The code of a VW can be accessed from a variety of devices, and it persists on the servers of service providers. Thus, according to them, this quality of code makes it analogous to physical objects.⁶⁷

However, this code can be accessed and modified at any given time by the developer, which is a critical weakness of this argument. Similarly, Erlank notes that permanence depends only on the cooperation of developers, who can make VP disappear at any time.⁶⁸ Chein warns that VWs are ephemeral and dynamic environments, and VP can be lost ‘at the accidental flick of a power switch’.⁶⁹ Cifrino also notes this potential obsolescence of VW business models, giving the example of *City of Heroes*, a VW which ceased operation in 2012, after eight years.⁷⁰

An issue related to the potential disappearance of VWs is the lack of interoperability between software in different VWs.⁷¹ So, when a user’s account has been restricted or terminated by one developer, the user cannot move it to another platform. There have been some efforts to make property in one VW compatible with the software in another VW, but until this is implemented, the quality of permanence remains rather dubious.⁷²

Lastowka and Hunter claim that temporality is a weak argument against virtual property. They use the examples of lease or usufruct, which are property interests recognised in common law and are time-limited. Due to its time-limited protection (seventy years post-mortem in the EU and US), IP is another example of rejecting the objection of temporality.⁷³ Therefore, the lack of permanence in the second and third layer VAs could serve as a solid argument against VP in the classical conception of property and its

⁶⁵ Castronova, ‘On virtual economies’ (n 5) 6.

⁶⁶ Fairfield, ‘Virtual property’ (n 7) 1054.

⁶⁷ Westbrook (n 7) 782–3.

⁶⁸ Erlank (n 1) 275.

⁶⁹ Chein (n 7) 1077.

⁷⁰ Cifrino (n 7) 23–4.

⁷¹ Glushko (n 8) 512; Nelson, ‘The virtual property problem’ (n 7).

⁷² For more about these efforts, see Vacca (n 7) 22; D Terdiman, ‘Tech titans seek virtual-world interoperability’ *CNET News* (15 October 2007) <http://news.cnet.com/Tech-titans-seek-virtual-world-interoperability/2100-1043_3-6213148.html> last accessed 1 August 2021 (noting the status of converting VWs to interoperability); Virtual World Interoperability <<http://VirtualWorldinterop.wikidot.com/start>> last accessed 1 August 2021 (summarising the results of the 2007 Virtual Worlds Interoperability Community Summit).

⁷³ Lastowka and Hunter, ‘Virtual worlds’ (n 1) 55–6.

permanence, and if one does not conceive of IP as property. On the other hand, notwithstanding the temporality of IP, the lack of this quality might exclude virtual property conceived in the traditional property sense but does not necessarily exclude proposing some other proprietary models for protecting virtual assets (like IP).

Fairfield argues that another physical quality of VW code is interconnectivity, analogous to the characteristic of objects in the real world (players can experience the connected world and interact with each other and the VW). Like Castronova,⁷⁴ Fairfield argues that 'code can be made interconnected, so that although one person may control it, others may experience it'.⁷⁵ As Erlank notes, if there were no interconnectivity in VWs, players would be able to experience only their own property, contrary to the whole idea of VWs.⁷⁶

However, code is not necessarily interconnected since not all computer systems can run all code without necessary adjustments, and we have the problem of interoperability, as seen in the discussion on permanence in the section above.⁷⁷

(b) Justifications of Virtual Property

The doctrinal question, that is, whether there *is* property in VWs, has been answered in the analysis above. VP does not share property concept incidents, and VAs do not possess characteristics of property objects. The following section will answer a normative question: whether there *ought to be* property in virtual worlds.

The key in recognising something as property is to identify the relevant theoretical justifications.⁷⁸ The next section will, therefore, refer to the leading Western justifications of *propertisation* (labour theory, utilitarianism and personhood theory), discussing the potential of their application to VAs and VWs. The analysis will use the layer classification explained above.

Labour Theory

Many authors contend that Locke's labour theory applies to virtual property. The main argument is that the time and effort users put in while creating

⁷⁴ Castronova, 'On virtual economies' (n 5).

⁷⁵ Fairfield, 'Virtual property' (n 7) 1054.

⁷⁶ Erlank (n 1) 246.

⁷⁷ Nelson, 'Fiber optic foxes' (n 7) 17.

⁷⁸ Erlank argues that virtual property could be more easily recognised in common law systems, as these 'just require a good justification'. See Erlank (n 1) 252.

virtual assets should entitle them to claim property rights.⁷⁹ The time players spend in VWs is significant if we look at the empirical research conducted on this topic.

Relevant research has been ambiguous. Nevertheless, it could be argued that the time players spend in VWs is quite considerable. For instance, in 2020, research showed that online video games were in the UK's top ten online activities.⁸⁰ In 2010, Mayer-Schönberger and Crowley asserted that 9.4 million players are each 'in-world' for about 22 hours per week, claiming that 'subscribers to VWs could be devoting over 213 million hours per week to building their virtual lives'.⁸¹ A global survey of 4,500 individuals in 2019 showed that those who play VWs or video games spend an average of 7 hours and 7 minutes each week playing and that the playing time had increased 19.3 per cent since 2018.⁸² MMORPGs are still in the top seven types of games played. The research also demonstrates a significant amount of time spent watching others play games on platforms such as Twitch.

On the first obvious question of whether we could consider 'game playing' as labour, it is argued that labour in the form of 'grinding' can be deemed relevant for labour theory. Grinding is a series of repetitive menial actions in VWs, completed to level up one's character.⁸³ In addition, the quality of labour can be demonstrated by looking at the phenomenon of 'gold farming'. Gold farmers are a particular subset of users who dedicate their hours 'in game' specifically to creating assets of value for later sale on in-game or grey markets.⁸⁴ Gold farms or 'gaming workshops' might employ a few dozen such farmers who perform various tasks specific to a particular game to build up virtual currency for the farm owners.⁸⁵ Although the data is somewhat uneven, there were some quite staggering estimates of the value of this 'virtual

⁷⁹ Shikowitz (n 8) 1015–54.

⁸⁰ Statista, 'Internet activities performed in Great Britain (UK) 2020' (10 November 2020) <<https://www.statista.com/statistics/275805/internet-activities-performed-in-great-britain/>> last accessed 1 August 2021.

⁸¹ Mayer-Schoenberger and Crowley (n 6) 1787; H Mahmassani et al., 'Time to play? Activity engagement in multiplayer online role-playing games' (2010) *Journal of the Transportation Research Board* 129–37.

⁸² Limelight Network, 'The state of online gaming 2019' <<https://www.limelight.com/resources/white-paper/state-of-online-gaming-2019/>> last accessed 1 August 2021.

⁸³ AE Jankowich, 'Property and democracy in virtual worlds' (2005) 11 *Boston University Journal of Science and Technology Law* 173, 183.

⁸⁴ R Heeks, 'Understanding "gold farming" and real-money trading as the intersection of real and virtual economies' (2010) 2(4) *Virtual Economies, Virtual Goods and Service Delivery in Virtual Worlds* 1, 6.

⁸⁵ Ibid. 7.

economy'. Heeks, for instance, estimated in 2010 that approximately 400,000 people were employed in gold farming, of which perhaps 85 per cent were based in China. In 1999, Ryan estimated that one million gold farmers were working on a global trade worth more than \$10 billion.⁸⁶ Therefore, the labour is already recognised as such in these black or grey markets. A more recent study demonstrates fluctuations in the virtual economy. According to the authors, the value of virtual currency in *World of Warcraft* 'plummeted heavily against real-world currencies across the world after the release of the game's newest expansion pack in August 2016'.⁸⁷

The argument against applying labour theory to VWs is that most players play games for entertainment purposes and not gold farming or labouring in general. Therefore, the time spent playing a game cannot qualify as adequate labour for labour theory.⁸⁸ Erlank replies to this objection by noting that not all VWs are used for entertainment (some are indeed used for many other purposes, including education, business and politics), and that the real world also rewards individuals who play games there, giving the example of athletes (professional ones are paid). Second, he comments that some players do indeed 'labour' by 'painstakingly' repeating the same actions in order to reap a reward, like blacksmiths for instance (referring to grinding mentioned above).⁸⁹

Advocates of applying Locke's theory to VP also argue that it is relatively easy to satisfy Locke's 'enough and as good' proviso in VWs (in short, that an individual can appropriate an object under the condition that there is enough and as good left for the others, the proviso subsequently revised by Locke⁹⁰). In VWs, arguably, there is an infinite number of resources available to players to labour and create.⁹¹ This does not have to be self-evident, however, as the abundance of the VW resources depends on the developers' will and actions, and for some of them, the users need to pay and do not labour on them (e.g. land in *Second Life*). The developers, therefore, artificially create a scarcity of resources in the VW. On the other hand, arguably, in-game resources are available to all players under the same conditions, and the developers

⁸⁶ Ibid.; ML Ryan, 'Immersion vs. interactivity: Virtual reality and literary theory' (1999) (28)2 *SubStance* 110.

⁸⁷ J Holm and E Mäkinen, 'The value of currency in World of Warcraft' (2018) *Journal of Internet Social Networking & Virtual Communities* article ID: 672253.

⁸⁸ Lastowka and Hunter, 'Virtual worlds' (n 1) 46; Erlank (n 1) 153.

⁸⁹ Erlank (n 1) 98.

⁹⁰ CB Macpherson, 'Editor's introduction' in J Locke *Second Treatise of Government: Essay Concerning the True Original Extent and End of Civil Government*, ed. with an intro. CB Macpherson (Hackett 1980) xvii.

⁹¹ Ibid. 64–5.

can adjust the scarcity feature, making more resources available if needed. Consequently, looking at a VW as a self-contained entity, this proviso seems fulfilled.

According to the proponents of applying labour theory to virtual property, Locke's spoilage proviso is also satisfied (the argument that the labourer is limited to 'as much as anyone can make use of to any advantage of life before it spoils'⁹²). The argument is that for the obvious reasons of the nature of VAs (underlying code that determines them), they cannot be spoilt, similar to money. Therefore, the limitation is unnecessary for VWs, since developers produce VAs and/or enable their creation by players and the limitation is embedded in the underlying code of the VW.

Lastowka and Hunter indicate that this justification for virtual property can be criticised based on Nozick's general objection to Locke's theory, namely, the labour which users embed in the VWs is insignificant compared with that of the owners of VWs.⁹³ Opponents of Nozick's argument argue that for some property, labour, no matter how insignificant it seems, adds value to a resource and recreates its essence.⁹⁴ Similarly, Lastowka and Hunter reply to this objection, arguing that this is correct in the sense that a player cannot claim property in the whole VW, but deserves property in the items where their labour makes up the most significant part of the value. They also assert that players do not claim property in the world itself but rather their items and avatars.⁹⁵

The most commonly articulated objection to applying Locke's theory to virtual property is the same one used against propertisation of IP, that is, the absence of commons.⁹⁶ According to this argument, the commons, the initial stage from which appropriation occurs, does not exist here, and VWs are not common *ab initio* but are usually owned by the developers. Therefore, according to labour theory, they seem to have a better claim as they actually invest their labour and resources in creating VWs.⁹⁷ Cifrino shares this stance, noting that if any labour on the initial commons would create property rights, borrowing and sharing of any object would be a problem if someone

⁹² Ibid. 60.

⁹³ Lastowka and Hunter, 'Virtual worlds' (n 1) 97; R Nozick, *Anarchy, State and Utopia* (Basil Blackwell 1974) 175.

⁹⁴ GS Alexander and EM Peñalver, *An Introduction to Property Theory* (Cambridge University Press 2012) 48.

⁹⁵ Lastowka and Hunter, 'Virtual worlds' (n 1) 63.

⁹⁶ See SV Shiffrin, 'The incentives argument for intellectual property protection' (2009) 4 *Journal of Law, Philosophy and Culture* 49, 96; RP Merges, *Justifying Intellectual Property* (Harvard University Press 2011) 35–9.

⁹⁷ Horowitz (n 61) 443–58.

later labours on that object and claims the title allegedly resulting from that labour.⁹⁸ Other authors reply to this, contending that the comparison could be made to Locke's commons created by God and VWs' commons created by their 'gods', or someone with godlike powers, the developers.⁹⁹ In addition, for those arguing that IP is property, in essence, the absence of commons can be bypassed and interpreted widely, as has happened practically.¹⁰⁰

Prima facie, labour theory presents a reasonable justification for recognising property in the second level VW's code, as this code satisfies the labour requirement and the two provisos (spoilage and 'enough and as good'). In addition, a player's labour constitutes the most significant part of the value of the VAs. For the first level items, understandably, the developer's labour and investment constitute the most significant part of their value; therefore, the developer should be entitled to own this layer.

However, the lack of the commons here is problematic as one cannot argue that there is any common of ideas, facts or resources in VWs.¹⁰¹ One way to neutralise this limitation would be to recognise the godlike powers of the developers and analogise them with God and Locke's commons. Alternatively, if the second layer is perceived separately and in relation to the other players and not the developer, then VWs features, which are open to all, can be seen as the commons. However, it is argued here that this argument is not plausible, mainly because labour put in by different individuals does not change the entire world, and the first layer remains developer-owned. Even if, arguably, there is a radical change in some instances (e.g. *Second Life* and similar worlds and games, where users do change the landscape significantly), this change does not defeat the property in the first layer. Locke's theory here thus serves better the interests of developers.

Personhood Theory

Personhood theories originate from Hegel's conception of property as an extension of personality,¹⁰² and Radin's classifications of property as fungible and personal. For Radin, property is an essential vehicle for developing the personality, and therefore, property that is especially close to a

⁹⁸ Cifrino (n 7).

⁹⁹ Erlank (n 1) 156–7.

¹⁰⁰ See for example J Peterson, 'Lockean property and literary works' (2008) 14(4) *Legal Theory* 257.

¹⁰¹ Apart from perhaps, open sources games, which are not the focus of this analysis.

¹⁰² GWF Hegel, *The Philosophy of Right*, trans. TM Knox (Oxford University Press 1967) para 41.

person's self-definition deserves special legal protections and precedence over fungible property.¹⁰³

This theory is, arguably, more applicable for justifying property interests in virtual assets than to justify traditional property.¹⁰⁴ In VWs, players are represented by a character, an avatar,¹⁰⁵ which is essentially a player's agent for interacting with the environment.¹⁰⁶ An avatar, and consequently a player, generally leads a more or less complete, rich and exciting life in VWs, often as a simulation of the real world. Using their avatars and offline, players communicate and socialise with others in the real world and gain reputation and social capital.

In most VWs, players usually establish substantial ties with their avatars, conceiving them as extensions of themselves, their psychological embodiments, *alter egos*.¹⁰⁷ A large body of research on VWs confirms this, referring to the concept of immersion.¹⁰⁸ Bartle, for instance, argues that VWs are all about 'the celebration of identity' and summarises the path players follow in the game in the phrase 'locate to discover to apply to internalise'. As the player develops, they travel from acquiring the skills for achieving something in the world, whatever that goal is, to exploring the world and applying the skills. The journey terminates with internalising the world and with complete immersion in it.¹⁰⁹ The concept of immersion in VWs is tied to presence and

¹⁰³ MJ Radin, 'Property and personhood' (1982) 34 *Stanford Law Review* 957.

¹⁰⁴ Lastowka and Hunter, 'Virtual worlds' (n 1).

¹⁰⁵ Castronova, 'On virtual economies' (n 5).

¹⁰⁶ For more about avatars, history and use in VWs, see Lastowka (n 17) 45–6; Dibble (n 3).

¹⁰⁷ Lastowka (n 17) 46; D Williams, T Kennedy and R Moore, 'Behind the avatar: The patterns, practices and functions of role playing in MMOs' (2011) *Games & Culture* 171.

¹⁰⁸ Y Lee and A Chen, 'Usability design and psychological ownership of a virtual world' (2011) 28(3) *Journal of Management Information Systems* 269; DA Bowman and RP McMahan, 'Virtual reality: How much immersion is enough?' (2007) 40(7) *Computer* 36–43; Ryan (n 86) 110; K Cheng and PA Cairns, 'Behaviour, realism and immersion in games' (in *Proceedings of the 2005 Conference on Human Factors in Computing Systems*, CHI 2005, Portland, Oregon, 2–7 April 2005 <<http://www.uecl.ac.uk/paul/research/Cheng.pdf>> last accessed 1 August 2021; L Ermi and F Mayra, 'Fundamental components of the gameplay experience: Analysing immersion' in S de Castell and J Jenson (eds), *Changing Views: Worlds in Play: Selected Papers of the 2005 Digital Games Research Association's Second International Conference*, 16–20 June 2005, Vancouver (DiGRA 2005) <http://www.uta.fi/~tlilma/gameplay_experience.pdf> last accessed 1 August 2021; P Sweetser and P Wyeth, 'GameFlow: A model for evaluating player enjoyment in games' (2005) 3(3) *ACM Computers in Entertainment* article 3A <http://www.itee.uq.edu.au/~penny/_papers/Sweetser-CIE.pdf> last accessed 1 August 2021.

¹⁰⁹ RA Bartle, 'Virtual worlds: Why people play' in T Alexander (ed.), *Massively Multiplayer Game Development 2* (Charles River Media 2005) 3–18 <<http://www.mud.co.uk/richard/VIRTUALWORLDWPP.pdf>> last accessed 1 August 2021; R Bartle, 'Presence and

an illusion that this computer-mediated environment is not mediated but real.¹¹⁰ The result of this ‘hill-climbing activity through identity space’ is ‘that players understand themselves more’.¹¹¹ Similarly, Lastowka shows immersion using the example of the use of language and the pronoun ‘you’ when referring to another person’s avatar and ‘I’ for one’s own avatar’s actions.¹¹²

The argument against using this theory to justify VP is found in the inalienability of personal property, as suggested by Radin and achieved, for instance, in the case of moral rights on the Continent.¹¹³ Therefore, this would result in proclaiming avatars and other second level VAs inalienable since they are so intrinsically related to a person. This is undesirable, however, since some users in VWs do want to trade their avatars, which often reach a considerable price on the markets.¹¹⁴ Lastowka and Hunter maintain that even if avatars and other second level VAs were to be proclaimed inalienable, on the practical side it is not a certain outcome as the courts might conclude otherwise and permit virtual trade.¹¹⁵ In addition, if classified as personal property, virtual assets would be better protected than fungible property, that is, the developers’ property, raising more disputes rather than providing solutions.¹¹⁶ On the other hand, something’s being deemed non-transferable does not necessarily exclude its proprietary character (e.g. common, public property).

An objection to this theory in general, and its application to virtual assets in particular, can be found in the argument of ‘separability’ or ‘thinghood’, meaning that things, in order to be property, must not be conceived of as ‘an aspect of ourselves or our on-going personality-rich relationships to others’ (e.g. blood, body parts, personal data).¹¹⁷ This objection is particularly applicable to avatars as property, considering the rich relationship between players

flow: Ill-fitting clothes for virtual worlds’ (2007) 10(3) *Techné: Research in Philosophy and Technology* 39.

¹¹⁰ M Lombard and T Ditton, ‘At the heart of it all: The concept of presence’ (2004) 3(2) *Journal of Computer-Mediated Communication* <<http://onlinelibrary.wiley.com/doi/10.1111/j.1083-6101.1997.tb00072.x/full>> last accessed 1 August 2021.

¹¹¹ Ibid. 15.

¹¹² Lastowka (n 17) 46.

¹¹³ See J Hughes, ‘The philosophy of intellectual property’ (1988) 77 *Georgetown Law Journal* 287; B Hugenholtz, ‘Copyright and freedom of expression in Europe’ in N Elkin-Koren and N Netanel, *The Commodification of Information* (Kluwer Law International 2002) 239, 240–1; P Rigamonti, ‘The conceptual transformation of moral rights’ (2007) 55 *American Journal of Comparative Law* 67–122, 98.

¹¹⁴ Ciffrino (n 7) 16; Lastowka (n 17) 176–7.

¹¹⁵ Lastowka and Hunter, ‘Virtual worlds’ (n 1) 65–6.

¹¹⁶ Erlank (n 1) 175–7.

¹¹⁷ See J Penner, *The Idea of Property in Law* (Oxford University Press 1997) 126.

and their avatars, but is less applicable to the other VWs items (swords, castles, houses, etc.).

Personhood theories could serve as a sound basis for justifying VP in the second and third level of code in VWs, those closely related to the player's personality, items and creations. The application of this theory, as demonstrated above, is not without difficulties and would not always serve players' interests (e.g. sale of avatars and other virtual assets).

Utilitarian Theory

Amongst the theories used in this chapter, the utilitarian theory is least applicable to justify virtual property in the second layer virtual assets. The main problem lies in the usefulness of VP for society, real-world non-players. It would potentially conflict with the felicific calculus principle of utilitarianism, looking for 'the greatest good for the greatest number'.¹¹⁸ Lastowka and Hunter would disagree with this assertion, claiming that in-game assets, from the utilitarian perspective, do not need to be beneficial for society; however, they are undoubtedly helpful and valuable for the individuals engaging in creating and improving these assets. Therefore, if the society (VW) is perceived as an aggregation of individuals (players), the utilitarian concept could be used. According to these views, by recognising VP, users would be rewarded for their efforts and incentivised to create further and develop VWs.¹¹⁹ An example of this is the exponential growth of *Second Life* users after Linden Lab changed its terms of service and promised ownership over players' creations.¹²⁰

On the other hand, players are already incentivised to create, and one of the major factors why they choose to join a particular VW is creation; therefore, property in VAs would probably not make much difference. Being in VWs already potentially results in economic benefits for players. Players can exchange their VAs for real money in many VWs, which is known as real money trading (RMT). RMT includes two main components: primary, which occurs within the game and under the end user licence agreement

¹¹⁸ J Bentham, *An Introduction to the Principles of Morals and Legislation*, ed. JH Burns and HLA Hart (Athlone Press 1970) 12–13.

¹¹⁹ See WM Landes and RA Posner, *The Economic Structure of Intellectual Property Law* (Harvard University Press 2003); WM Landes and RA Posner, 'An economic analysis of copyright law' (1989) 18 *Journal of Legal Studies* 325, 326; S Leung, 'The commons and anticommons in intellectual property' (2010) *UCL Journal of Law and Jurisprudence* 16; S Kieff, 'Property rights and property rules for commercializing inventions' (2001) 85 *Minnesota Law Review* 697; EW Kitch, 'The law and economics of rights in valuable information' (1980) 9 *Journal of Legal Studies* 683.

¹²⁰ Vacca (n 7).

(EULA),¹²¹ and secondary, which happens outside the game and beyond the EULA provisions. Players usually make money from selling VAs on online auctions within or outside the VW (some of the VWs expressly ban the use of external auctions, e.g. Blizzard's *World of Warcraft* EULA).¹²² For instance, in 2006, Anshe Chung accumulated more than one million dollars in virtual assets, becoming the first millionaire of the popular VW *Second Life*.¹²³ In December 2009, a person known as 'Buss Erik Lightyear' paid \$330,000 to own a virtual space station in *Planet Calypso*, an MMORPG.¹²⁴ The game allows exchanges between virtual currency and real dollars at a fixed exchange rate of 10 PED (virtual currency) to \$1.¹²⁵ Wu estimates that the market for virtual goods in the US exceeded \$3 billion in 2012 and it 'is expected to grow briskly in later years'.¹²⁶ In 2013, Linden Lab reported 1.2 million daily transactions for virtual goods and \$3.2 billion worth of transactions in *Second Life* Economy.¹²⁷ However, it is still unclear whether there could be a further explosion in the numbers of VWs users and their transactions, provided that VP is recognised.

The incentives argument, therefore, works much better for developers. Creating and maintaining a VW can be a profitable business deal as they can earn from various sources, such as subscriptions, virtual sale commission,

¹²¹ 'The EULA is a software license between the developer and the user (and generally drafted by the developer) that governs the relationship between these two parties. The EULA is generally presented as a graphical computer window that pops up when the [user] of the software begins running the program. The [user] is then presented with the terms of the license, and must click a button indicating that he/she has read and accepted those terms.' The software will only begin running if the user agrees to the EULA. See *ibid.* 43.

¹²² An example of the existing auctions site is <<https://www.playerauctions.com/wow-classic-tbc-items/>> last accessed 1 August 2021.

¹²³ R Hof, 'Second Life's first millionaire' *Businessweek Online* (2006) <<https://www.bloomberg.com/news/articles/2006-11-25/second-lifes-first-millionaire>> last accessed 1 August 2021.

¹²⁴ M Schramm, 'Man buys virtual space station for 330k real dollars' *Joystiq* (2010) <<http://www.joystiq.com/2010/01/02/man-buys-virtual-space-station-for-330k-real-dollars/>> last accessed 1 August 2021.

¹²⁵ S Brennan, 'Crystal Palace space station auction tops 330,000 US dollars' *Joystiq* (2009) <<http://massively.joystiq.com/2009/12/29/crystal-palace-space-station-auction-tops-330-000-us-dollars/>> last accessed 1 August 2021.

¹²⁶ S Wu, 'Digital afterlife: What happens to your data when you die?' (2013) <<http://dataedge.ischool.berkeley.edu/2013/pdf/digital-afterlife-white-paper.pdf>> last accessed 1 August 2021.

¹²⁷ See *Second Life*, 'Second Life affiliate program' <<http://secondlife.com/corporate/affiliate/?lang=it-IT>> last accessed 1 August 2021.

purchase of land and other features.¹²⁸ In order to achieve this, understandably, they need to have a significant user base,¹²⁹ incentivised perhaps by VP rights. In addition, they need to have their rights in the first layer VAs to prevent free riding on their creations.

The free riding arguments (arguments against allowing an individual to obtain benefits from someone else's investment, preventing them from recouping costs) are also somewhat applicable to the second layer too, in the sense that VWs as a society take advantage of and become more attractive to new users with these creations, which developers profit from.¹³⁰ Another likely scenario is the free riding of other players, replicating and copying other players' creations (e.g. their original swords, houses, ships). However, as noted by Lemley for IP rights, free riding might even be desirable in VWs, as there is much less need to internalise negative externalities. As with IP, negative externalities are less prominent here than tangible property, as consumption by many players is desirable since it enriches the society and culture of VWs.¹³¹ Also, the lack of scarcity in VWs means that free riding would not result in severe detriment, and the developers could make more resources available to players.

Conversely, one of the arguments *contra* using this justification for VP rights is the allocation reason. According to this view, utilitarian theories could be used to oppose the creation of property rights in VWs. They would decrease the welfare of VWs' owners and other users by giving property to individuals and creating, effectively, *the tragedy of anticommons*, where individuals would be able to prevent the use of VP, resulting in unwanted underuse

¹²⁸ For Blizzard profits, see for example E Makuch, 'Activision Blizzard profits hit \$1.1 billion in 2012' *Gamespot* (2013) <<http://www.gamespot.com/articles/activision-blizzard-profits-hit-11-billion-in-2012/1100-6403613/>> last accessed 1 August 2021. For Linden Lab's, see J Reahard, 'Linden Lab's Second Life "extremely profitable" company looking to expand' *Massively by Joystiq* (2012) <<http://massively.joystiq.com/2012/03/15/linden-labs-second-life-extremely-profitable-company-looking/>> last accessed 1 August 2021.

¹²⁹ Statistics about the MMORPGs market show that in 2013 there were approximately 20 million subscribers, and that the peak in terms of numbers of subscribers was in 2011, close to 23 million. See TotalSubs, MMODData.net <<http://users.telenet.be/mmodata/Charts/TotalSubs.png>> last accessed 1 August 2021. The latest data still reports millions of *WoW* and other MMORPGs users worldwide (4.7 million). See Statista, 'In 2015, when Activision Blizzard last reported on World of Warcraft's subscriptions, the game had 5.5 million global subscribers. It is projected that the numbers will gradually decrease to hit 4.46 million in 2023' <<https://www.statista.com/statistics/276601/number-of-world-of-warcraft-subscribers-by-quarter/>> last accessed 1 August 2021.

¹³⁰ Landes and Posner, 'An economic analysis of copyright law' (n 119); MA Lemley, 'Property, intellectual property, and free riding' (2005) 83 *Texas Law Review* 1031, 1059–60.

¹³¹ Lemley, 'Property' (n 130) 1059–60.

of VWs by players.¹³² Lastowka and Hunter reply to these arguments, saying that they do not consider justifications for allocation but rather for creating property rights in virtual goods. According to them, this does not mean that property should not exist in VWs at all. The argument states that property is not allocated correctly, which can be corrected by the courts, for instance.¹³³ This response does not address the objection adequately. Instead, the nature of VWs and the layer approach would prevent underuse, as the first layer belongs to the developers and the rights in the second one are derived from this ownership.

5.5 Allocation of Ownership in Virtual Worlds

In this section, we assess the allocation of property in VWs, arguing that most developers curtail possibilities for players to assert any VP rights in second level VAs, even if theories of property justify this concept.¹³⁴ Moreover, where developers have envisaged some kind of player property rights in their EULAs (e.g. *Second Life*), these rights are minimal and can barely be categorised as property. The solution to rectify this imbalance, therefore, is potentially available in the form of consumer protection. However, due to the distinctive character of VWs and the areas these contracts aim to regulate, consumer protection laws do not prove very helpful.

Allocation of ownership, IP and other different rights in VWs is established through contracts. VW contracts come in the form of clickwrap licences (end user licence agreements – EULAs, terms of service – ToS, rules of conduct and other different policies).¹³⁵ The effects of these contracts are widely disputed, as they leave little or no freedom for the user, and no other choice apart from clicking ‘I agree’ or to decline, therefore refusing to take part in the game.¹³⁶ The most common model is that the developer claims all property and IP rights associated with the VW.¹³⁷

¹³² See MA Heller, ‘The tragedy of the anticommons: Property in the transition from Marx to markets reviewed’ (1998) 111(3) *Harvard Law Review* 621.

¹³³ Lastowka and Hunter, ‘Virtual worlds’ (n 1) 59–60.

¹³⁴ Harbinja, ‘Virtual Worlds – a legal post-mortem account’ (n 12).

¹³⁵ See for example Blizzard, ‘World of Warcraft – end user license agreement’ <<https://www.blizzard.com/en-us/legal/fba4d00f-c7e4-4883-b8b9-1b4500a402ea/blizzard-end-user-license-agreement>> last accessed 1 August 2021.

¹³⁶ Erlank (n 1); T Pistorius, ‘Click-wrap and web-wrap agreements’ (2004) 16 *SA Mercantile Law Journal* 568–76; MA Lemley, ‘Terms of use’ (2006) 91 *Minnesota Law Review* 459, 459–83.

¹³⁷ Jankowich conducted a survey of forty-eight VWs, confirming these assertions. See Jankowich, ‘Property and democracy’ (n 83).

Blizzard, the *World of Warcraft* and *Diablo III* developer, expressly excludes any property rights of users in assets created or traded in the game, in addition to forbidding the transfer of accounts (s 1 B iii of Blizzard EULA). Conversely, *Second Life* and Linden Lab give relatively extensive rights in content created by users. Initially, Linden labelled these rights as property, but in response to *Bragg v Linden Research, Inc.*,¹³⁸ changed its terms to granting an IP right only.¹³⁹ It also denies property rights in virtual currency (Linden dollars), and land that users can buy in *Second Life*, reminding them of the limited licence they are granted.¹⁴⁰ Moringiello argues that Linden deceives its users as it effectively promises something resembling the bundle of land rights, that is, property, and then takes it back by way of service.¹⁴¹ As Erlank rightly notes, even the recognised rights are somewhat illusory, as Linden limits them to the game and refuses any liability and compensation in the case of damage or loss of this property.¹⁴² Nevertheless, he also reasonably opines that by insisting on regulating and limiting VP, the developer implicitly recognises its existence.¹⁴³

On the other hand, Linden grants itself a non-exclusive licence in players' creations, the scope of which has been widened even more recently, leaving many players of *Second Life* displeased and embittered, wanting to leave.¹⁴⁴ Also, its EULAs caused Linden Lab to be involved in the most critical court cases about VWs and VP in the Western world.

¹³⁸ 487 F.Supp.2d 593, 612 (ED Pa 2007).

¹³⁹ Linden Lab, 'Second Life residents to own digital creations' *Press Release* (2003) <<http://creativecommons.org/press-releases/entry/3906>> last accessed 1 August 2021; 'Second Life terms of service' (15 December 2010) <<https://secondlife.com/app/tos/tos.php>> last accessed 1 August 2021, title 2; and see commentary in Vacca (n 7); AB Steinberg, 'For sale – one level 5 barbarian for 94,800 won: The international effects of virtual property and the legality of its ownership' (2009) 37 *Georgia Journal of International and Comparative Law* 381; J Gong, 'Defining and addressing virtual property in international treaties' (2011) 17 *Boston University Journal of Science and Technology Law* 101.

¹⁴⁰ See *Second Life*, 'Terms of Service' s 1.2 <<http://lindenlab.com/tos>> last accessed 1 August 2021.

¹⁴¹ Moringiello (n 7).

¹⁴² *Second Life* ToS part 9, or part XVII *WoW* EU ToS, or part 12 US. See Erlank (n 1) 102.

¹⁴³ Erlank (n 1) 112.

¹⁴⁴ M Korolov, 'Outrage grows over new Second Life terms' *Hypergrid Business* (30 September 2013) <<http://www.hypergridbusiness.com/2013/09/outrage-grows-over-new-second-life-terms/>> last accessed 1 August 2021; 'SecondLife® Content Creators Survey on Linden Lab TOS issue' <<http://toytalks.weebly.com/1/archives/09-2013/1.html>> last accessed 1 August 2021.

The case of *Bragg v Linden Research, Inc.*¹⁴⁵ was introduced in the above section 'Bragg Case: Property and Jurisdiction in VWs'. As noted there, the court confirmed that Linden's terms of service were unconscionable in relation to the mandatory arbitration clause and knocked the clause down. Importantly, the court said that the terms left the plaintiff with no effective remedy.¹⁴⁶

More recently, in *Evans et al. v Linden Research, Inc et al.*, once again, there was no decision regarding virtual property. The case was settled, and even its relevance to the validity of the EULA is limited.

Even the 'liberal' VWs and games seem to be replicating these EULAs. An example of this is Steam, an entertainment platform distributing different games, including VWs. This very successful platform is considered user-friendly, open-source to an extent, and an alternative to the traditional business models.¹⁴⁷ Valve, the owner of Steam, created a very restrictive EULA (subscriber agreement), very much resembling those of other VWs. Therefore, apart from IP rights,¹⁴⁸ players' ownership over their creations and virtual money contained in their wallets¹⁴⁹ is limited and non-transferable, with an extended licence taken by the provider, Valve Corporation.¹⁵⁰ Valve has been criticised for banning a user who, contrary to the EULA, attempted to sell his Steam account.¹⁵¹

Following the above analysis, it could be argued, as many authors do, that the contracts are *prima facie* unfair.¹⁵² The reasonable remedy would be challenging their unfair or unconscionable provisions in courts using consumer protection law.¹⁵³ One could argue that consumer protection law might be helpful to resolve these issues. At the level of the EU, Directive 2011/83/EU

¹⁴⁵ 487 F.Supp.2d 593, 612 (ED Pa 2007).

¹⁴⁶ Hetcher (n 26) 829, 836.

¹⁴⁷ 'The platform's record was from May 2021, peaking at 26.85 million concurrent users. Steam had approximately 120 million monthly active players in 2020.' See Statista, 'Number of peak concurrent Steam users 2013–2021' <<https://www.statista.com/statistics/308330/number-stream-users/#:~:text=Steam%20had%20approximately%20120%20million,monthly%20active%20users%20in%202019>> last accessed 1 August 2021.

¹⁴⁸ Steam, 'Subscriber agreement' s 6 <http://store.steampowered.com/subscriber_agreement/> last accessed 1 August 2021.

¹⁴⁹ Ibid. part C.

¹⁵⁰ Ibid. s 6 A.

¹⁵¹ A Webster, 'Steam user violates subscriber agreement, loses \$1,800 in games' *Ars Technica* (2011) <<http://arstechnica.com/gaming/2011/03/steam-user-violates-subscriber-agreement-loses-1800-in-games/>> last accessed 1 August 2021.

¹⁵² Jankowich, 'Complex web of corporate rule-making' (n 8) 50.

¹⁵³ Riley (n 8) 907.

of the European Parliament and of the Council of 25 October 2011 on consumer rights would apply.¹⁵⁴ This Directive, implemented in the UK in the form of the Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013 SI No 3134 and the Consumer Rights Act 2015, encompasses contracts regarding digital content, including games (see Recital 19 of the Directive).¹⁵⁵ According to the Act, terms that would potentially be deemed invalid are, for example, those limiting the developer's liability, reserving the right to terminate or modify terms discretionary and without notice, and arbitration clauses.¹⁵⁶

Both the UK and EU legislation, however, apply to issues such as information to consumers, rights of withdrawal, liability, delivery and passing of risk, but do not address the issues of property rights, as subject matter cannot be considered unfair, and this is out of the scope of this legislation.¹⁵⁷ This law could apply to the parts of the contracts regulating the licence sale for using software (the first layer of VWs). However, the second and third layers are players' creations and would not fall within the definition of goods and services found in the consumer protection laws (as they are not goods or services sold by the developers). Similar, though much more limited protection can be found in California, mandated through the Consumers Legal Remedies Act (2006), including the prohibition of inclusion of previously discussed unconscionable provisions in the contract.¹⁵⁸

So far, VW contracts have not been challenged much in the US and UK courts. In the UK, there is no such case at the time of writing. Most case law regarding online games has considered IP aspects of the game features, disputes between the developers, but rarely those between users and service

¹⁵⁴ OJ L 304, 22/11/2011 0064–0088. This Directive replaces, as of 13 June 2014, Directive 97/7/EC on the protection of consumers in respect of distance contracts OJ L 144/19 and Directive 85/577/EEC to protect consumers in respect of contracts negotiated away from business premises OJ L 372/31.

¹⁵⁵ Part 2 of the Act replaces the Unfair Terms in Consumer Contracts Regulations 1999. The Consumer Rights Act 2015, c 5 implements Directive 2011/83/EU of the European Parliament and of the Council of 25 October 2011 on consumer rights, amending Council Directive 93/13/EEC and Directive 1999/44/EC of the European Parliament and of the Council and repealing Council Directive 85/577/EEC and Directive 97/7/EC of the European Parliament and of the Council OJ L 304/64, 22/11/2011.

¹⁵⁶ Schedule 2 of the Act lists non-exhaustively terms that might be regarded as unfair.

¹⁵⁷ See the Consumer Rights Act 2015, s 64. See also Recital 51 of Directive 2011/83/EU of the European Parliament and of the Council of 25 October 2011 on consumer rights. Also, the EU does not interfere with property rights of member states. See Article 345 of the Treaty on the Functioning of the European Union (Consolidated version 2012) OJ C 326, 26/10/2012.

¹⁵⁸ California Civil Code §§ 1750 *et seq.*

providers when it comes to property. The US case law is more developed, and the Bragg and Evans courts found specific contract provisions unfair (jurisdiction, account suspension). Nevertheless, the courts' deliberations on property rights have been entirely accidental in discussing the main legal issues. Therefore, we should not rely on court cases to come in and resolve the issue of virtual property any time soon. Even if more cases were to appear, the outcome in the US might not be beneficial for players.¹⁵⁹

To conclude, VW contracts, at the moment, deny players' virtual property rights in their creations and VW items. However, the courts have occasionally attempted to address the balance via doctrines of unfairness in contracts, which could be a potential solution. In principle, the question of creating and/or recognising proprietary rights and interests in VWs is not an issue that contracts can regulate but one of the general laws of property or IP. In addition, an attempt to apply consumer protection law to VWs' EULAs and allocation of property therein is contrary to the views of many authors mentioned in the subsequent section, namely, that VWs are not just games and their players are not just users but active participants, citizens, residents of the world. I too have argued this: I have analysed the constitutionalisation of VWs extensively in my earlier work and argued that VWs players possess characteristics of citizens of these worlds, and their statutes should be reconsidered accordingly.¹⁶⁰

I have also argued that the present form of regulation, by contracts and code, is inadequate for protecting users' interests, especially their interest in autonomy. Relationships between players and providers often have arbitrary and *ad hoc* outcomes.¹⁶¹ While this balance of power may promote commercial certainty and predictability for developers and significantly lower costs for users, the status quo is still unsatisfactory. The quasi-constitutional relationships are unfair and unsuitable, and there is a need for more accountability of service providers to their users. Recognising the features of VWs,

¹⁵⁹ See Gilbert (n 6) 242. Generally, see S Randall, 'Judicial attitudes toward arbitration and the resurgence of unconscionability' (2004) 52 *Buffalo Law Review* 185. On the unconscionability and Californian law, see Quinn (n 8). See also Chein (n 7).

¹⁶⁰ Harbinja, 'Virtual worlds – a legal post-mortem account' (n 12).

¹⁶¹ Jankowich, 'Complex web of corporate rule-making' (n 8); Lastowka (n 17); Erlank (n 1); Fairfield, 'Virtual property' (n 7); Castronova, 'On virtual economies' (n 5); Castronova, 'Real products in imaginary worlds' (n 5); Castronova, 'The right to play' (n 5); Castronova, 'Virtual world economy' (n 5). I have followed their approach in Harbinja, 'Virtual worlds players – consumers or citizens?' (n 12).

their distinct character and place-like qualities, it is necessary to provide for a better legal and regulatory regime to protect their citizens.¹⁶²

Such a system would recognise a service provider's interest and property or IP in the system and the software, but it would also consider the user's autonomy and choice over what happens to their VAs on death. The solution suggested in the section below is based on concepts drawn from private law, not public law, but it does consider the features of VWs (immersion, place-like features) and recognises the *constitutional* nature of VWs.¹⁶³

5.6 Alternatives: Property Rights in the Property of Another

The analysis has so far been normative and theoretical with reference to the law. In the subsequent sections, however, the analysis will become more doctrinal to reflect legally on the specific nature of VWs. For the reasons identified when discussing virtual property above, it is argued here that full ownership of VP is not an adequate solution as it would be prejudicial to the interests of either the players or the developer of the VW. We therefore need more nuanced solutions that would serve as a compromise between these interests.

Some proposals have already experimented with property interests other than full ownership. These come in the form of limited real rights, derived from and subordinate to another person's full ownership. In civil law systems, these rights are known as servitudes: real (falling on immovable property) or personal (attached to a person, allowing them to enjoy the property of another).¹⁶⁴ In common law, these time-limited rights are usually only attached to immovables (real property) and are represented by notions such as easements, lease or life estate.¹⁶⁵ It is argued that such rights can serve usefully as models to consider the fact that the interests and rights of players are based on someone else's property (referred to above as the 'first layer', the developer's code and servers). Since VWs operate globally, these proposals need to experiment with both civil and common law concepts, trying to identify commonalities and to strike the best balance for VW players. I do not suggest in my earlier work that these common and civil law concepts should or can be merged and borrowed in either of the real-world jurisdictions, notwithstanding the

¹⁶² Lastowka (n 17); Castronova, 'On virtual economies' (n 5); Castronova, 'Real products in imaginary worlds' (n 5); Castronova, 'The right to play' (n 5); Castronova, 'Virtual world economy' (n 5).

¹⁶³ Harbinja, 'Virtual worlds players – consumers or citizens?' (n 12).

¹⁶⁴ Bell et al. (n 49) 289–90; E Steiner, *French Law: A Comparative Approach* (Oxford University Press 2010) 389–90.

¹⁶⁵ M Dixon, *Modern Land Law* (7th edn, Routledge 2010) 267–8, 313.

arguments for legal transplants and borrowing in general.¹⁶⁶ As explained above, the proposal is a reform proposal, limited to VWs as the separate, peculiar places we consider them to be.

(a) *Suggested Models: Virtual Easement*

Slaughter proposes an exciting model, analysing the benefits and drawbacks of introducing a property or contractual regime for VWs. He introduces the concept of 'virtual easement'. According to him, this servitude would feature transferability (from one user to another, in life and on death); longevity (for as long as the user invests time and/or money and the VW exists); liability (no property remedies); *in rem* nature (except for the liability rule, which is *in personam*); and *numerus clausus* (finite number of iterations).¹⁶⁷ This theory appears as rather original and a good compromise between the rights of users and those of service providers. However, the flexibility it offers could be perceived as a possible source of uncertainty for players since different service providers could choose different terms also to their detriment, which is usually not the case with servitudes in the real world, especially in civil law systems, where the certainty of property rights is considered as an ultimate aim.¹⁶⁸

Similarly, the system of easements (the common law counterpart of the civil law servitudes) has been argued for by Lastowka in his later work.¹⁶⁹ He sees it as the best solution since both the players and the VW owners are interested in something that depends on essentially one tangible thing, namely servers owned by the providers. Therefore, to enable rights on top of this ownership interest, it is necessary to introduce lesser rights for the benefit of VW inhabitants.¹⁷⁰ He does not suggest what features this model could have. A similar solution was offered by Fairfield in his later work.¹⁷¹ Under the model he proposes, the licence agreement would also recognise covenant-style interests or servitude of users.¹⁷² The problem with easements, covenants and leaseholds would be that, by definition, these interests are related to land,

¹⁶⁶ Harbinja, 'Virtual worlds – a legal post-mortem account' (n 12).

¹⁶⁷ Slaughter (n 7).

¹⁶⁸ This principle permeates legal writings referring to civilian systems, and their mandatory rules for property and rigidity, as van Erp notes: 'As a result, property law became a rather petrified legal area, rooted in a desire for legal certainty.' See S van Erp, 'Comparative property law' in M Reimann and R Zimmermann (eds), *The Oxford Handbook of Comparative Law* (Oxford University Press 2006) 1044.

¹⁶⁹ Lastowka (n 17) 127.

¹⁷⁰ *Ibid.*

¹⁷¹ Fairfield, 'Anti-social contracts' (n 8).

¹⁷² *Ibid.* 451–7.

immovable property.¹⁷³ In that case, we would have to use a somewhat weak analogy between land and developers' server systems.

(b) Suggested Models: Intangible Usufruct

Veloso introduces the concept of 'intangible usufruct'. He asserts that this is a good solution for the practical reason that it avoids one-sided arguments and aims to provide a way out of unfair contracts, while still respecting developers' interests.¹⁷⁴ He proposes three rules to govern the relations established by usufruct. First, that the developer should be considered the owner, under contract, and should provide for the right to use and the right to the fruits for the user; these rights are alienable and, when bundled together, should form a virtual property right.¹⁷⁵ Second, the developer may undertake any works and improvements or diminution on virtual property and/or the VW, but provided that such acts are not exercised arbitrarily, should they cause a diminution in the value of the usufruct or prejudice the right of the user.¹⁷⁶ Third, if the VW is terminated, the players are considered to have returned their virtual property to the developer thereby absolving the developer from any complaint that might arise. This approach appears reasonable, and the solution in this chapter will build upon this proposal, developing it in more detail, especially concerning the transmissibility and considering the different conceptions of servitudes (usufruct) between legal systems. I introduced the proposal in my earlier work, albeit tentatively and in less detail than in this book.¹⁷⁷

(c) Proposal: VWs Usufruct

This chapter puts forward a reform proposal designed to balance the interests of the creators of VWs and the players or users in these worlds. Though not conceptually aligned with civilian usufruct and common law life estate (for reasons discussed in detail below), this model is inspired by it.¹⁷⁸

As noted above, Veloso, Slaughter, Lastowka and I, in my previous work, suggest solutions in this domain modelled on civilian usufruct and common

¹⁷³ See for example C van der Merwe and AL Verbeke (eds), *Time Limited Interests in Land* (Cambridge University Press 2012); Restatement (Third) of Property (Servitudes) (2000) §§ 4.6, 1.1 and 1.2.

¹⁷⁴ Veloso (n 7).

¹⁷⁵ Ibid. 73.

¹⁷⁶ Ibid. 74.

¹⁷⁷ Harbinja, 'Virtual worlds – a legal post-mortem account' (n 12).

¹⁷⁸ Ibid.

law easement. Therefore, it is essential to review the implications of the models based on usufruct before introducing the new reform proposal.

Usufruct

Usufruct is a civil law concept that originates from Roman law, and it essentially entitles a person to the rights of use and fruits of another person's property. Under Roman law, the usufructuary had the right to use and enjoy the property and its fruits while preserving the substance of the property (i.e. the elements of *usus* – use, and *fructus* – fruits of one's property, but lacking *abusus* – alienation and transmission).¹⁷⁹ The common law concept of life estate has similar effects in relation to rights conferred to a life-tenant, particularly regarding the enjoyment of fruits.¹⁸⁰ In Scots law, a mixed legal system, a similar role is played by liferent. Usufruct does not have to pertain to immovable property; it can be created over movable and immovable property, although it is more commonly attached to the land.¹⁸¹ The property owner retains nude ownership, that is, ownership burdened with a real right of enjoyment and use. The French and Belgian Civil Codes (FCC and BCC) employ a similar description, namely *usufruit* and *vruchtgebruik* respectively (FCC/BCC, Article 578).¹⁸² In German law, the property may be similarly burdened with a *Niessbrauch* (BGB, §1030).

Usufruct is a real right, a right to the property itself and not merely a right against a person (the owner).¹⁸³ However, a key feature of usufruct is that it terminates on the death of the usufructuary (if not earlier).¹⁸⁴ This obvious temporal limitation does not prevent the right of usufruct from being alienated but only limits the time frame of the transferred usufruct to the life of

¹⁷⁹ Gaius, 2.30; D.7.1.1; D.6.1.33; D.7.1.72; D.7.4.2; D.23.37.8.3.S in P Scott (ed.), *Civil Law, Including The Twelve Tables, The Institutes of Gaius, The Rules of Ulpian, The Opinions of Paulus, The Enactments of Justinian, and The Constitutions of Leo* (Central Trust 1932) 286 <<http://www.constitution.org/sps/sps.htm>> last accessed 1 August 2021.

¹⁸⁰ J McClean, 'The common law life estate and the civil law usufruct: A comparative study' (1963) 12(2) *International & Comparative Law Quarterly* 649, 658.

¹⁸¹ *Ibid.*

¹⁸² The description in Old Dutch Civil Code, art. 803 (in force until 1992) was derived from Civil Code, art. 578. AL Verbeke, B Verdickt and DJ Maasland, 'The many faces of usufruct' in van der Merwe and Verbeke (n 173) 33–57, 37, citing C Asser, FHJ Mijnsen, AA van Velten and SE Bartels, *Mr. C. Asser's Handleiding tot de beoefening van het Nederlands burgerlijk recht 5, Zakenrecht, Eigendom en beperkte rechten* (5th edn, Kluwer 2008).

¹⁸³ Zenati-Castaing and Revet, *Les Biens*, 494 and de Page, *Traité élémentaire*, 153, cited in Verbeke et al. (n 182) 36.

¹⁸⁴ Cass. 3 July 1879; Pas. 1879 I 342; Borkowski and Du Plessis, *Roman Law*, 172; Baudry-Lacantinerie, *Précis de droit civil*, 770; de Page, *Traité élémentaire*, 258; Prutting, *Sachenrecht*, 364; Verbeke, 'Quasi-vruchtgebruik', 37, all cited in Verbeke and Maasland (n 182) 36.

the original usufructuary (FCC, Article 617). In principle, usufruct expires upon the death of the person on whose life the right was based, irrespective of any contracted term.¹⁸⁵ It is not possible to constitute a usufruct that is permanent or unlimited in time.¹⁸⁶ However, historically and nowadays, any fruits which the usufructuary has already gathered before death would pass to their heirs.¹⁸⁷ McClean finds that the common law concept of life estate and Scots law liferent have a similar effect to usufruct and confer almost the same rights.¹⁸⁸

There are some problems with applying a solution based on the usufruct concept to VWs. Importantly, as noted above, usufruct in principle ends on the death of the usufructuary. Hence, nothing would persist in being transmitted to heirs on that person's death. Although some recent legal reforms in civil law countries such as the Netherlands demonstrate that there can be modifications to the usual rules of usufruct in the interests of policy,¹⁸⁹ no such reforms have affected the fundamental idea that the usufruct interest terminates on the death of the usufructuary.

A second problem relates to applying a solution based on usufruct to common law or other legal systems. Similar concepts (easements, leasehold or life estate in England) only apply to immovable property (or 'real property' in English law) and have very different legal nature and effects (regarding duration, use, transfer, etc.). It is not clear whether life estate in common law (the concept resembling usufruct most) applies to movables or relates only to land and immovable property.¹⁹⁰ As noted above, McClean argues that there is no actual difference in substance between usufruct and life estate and that in a mixed legal system, such as Scottish liferent, the right to use another's property (the fee) for life can extend to movables.¹⁹¹ Notwithstanding these

¹⁸⁵ Verbeke et al. (n 182) 36n27.

¹⁸⁶ Ibid. 37n15. The temporal aspect also applies to the common law life interest. See McClean (n 180) 655; FH Lawson and B Rudden, *The Law of Property* (Oxford University Press 2002) 97.

¹⁸⁷ A Watson, *The Digest of Justinian, Volume 1* (University of Pennsylvania Press 1998) 242–3; W A Hunter, *A Systematic and Historical Exposition of Roman Law in the Order of a Code* (Sweet & Maxwell 1803) 388–9; Code civil, Articles 582–99.

¹⁸⁸ McClean (n 180).

¹⁸⁹ The Dutch Civil Code in Article 3:212, § 1 stipulates that if usufructuary assets are destined to be alienated, the usufructuary is entitled to alienate the assets in accordance with their intended purpose. See Verbeke et al. (n 182) 51–2.

¹⁹⁰ Ibid.

¹⁹¹ GL Gretton and AJM Steven, *Property, Trusts and Succession* (2nd edn, Bloomsbury Professional 2013) 323; WJ Dobie, *Manual of the Law of Liferent and Fee in Scotland* (W. Green & Son 1941) 1–2.

differences between legal systems, the global uptake of VWs by players from many different jurisdictions means a solution based purely or mostly on one of these usufructuary or similar institutions may not scope well to VWs.

While usufruct has been an exciting inspiration in its original and pure form, it cannot be the foundation of a successful, multi-jurisdiction solution for providing rights to players to bequeath the value of their second layer VAs to their heirs. I have justified the need for such a solution earlier in this chapter and in my previous research, discussing constitutionalisation, the pervasive and immersive nature of the VWs environment, and time and labour players employ in VWs.

VWs Usufruct

The new model I suggest is ‘VWs usufruct’, an entirely new right, which could be introduced by domestic statutes or model law as in the US. VWs usufruct is a personal right of a player against a VW provider in second layer assets. The player has the right to use and transfer their second layer asset while playing the game. In addition, the player has a right to compensation in the form of monetary value for assets they have earned, acquired or purchased in the game, and such compensation belongs solely to their account and applies to their second layer assets.

With limitations explained in this section, the VWs usufruct transmits to heirs on death, which differs from the model suggested in my earlier work. Users who ‘own’ or create virtual assets acquire only contractual rights against the VW’s owner through its ToS. Such personal contracts will be discharged on death unless there is an opposite provision in the contract.¹⁹² As contracts rather routinely expressly exclude survivability, the transmission of virtual assets on death is impossible under the current VWs’ ToS.

The right to transmit the VW’s usufruct will only apply to the monetised value of assets that can be transferred. This differs from the only detailed post-mortem related proposal in VWs scholarship, introduced by Troung.¹⁹³

¹⁹² Principle ‘*Actio personalis moritur cum persona*’ in *Baker v Bolton* (1808) 1 Camp 439; 170 ER 1033, revised by the Law Reform (Miscellaneous Provisions) Act 1934, c 41 (as amended), revised the rule mandating that all personal rights will survive against and for the benefit of the estate, with the only exception of defamation and claim for bereavement. For a commentary about the contracts and succession, see AR Mellows, *The Law of Succession* (4th edn, Butterworths 1983) 295–6; B Nicholas, *The French Law of Contract* (2nd edn, Clarendon Press; Oxford University Press 1992) 29–30; ML Levillard, ‘France’ in DJ Hayton (ed.), *European Succession Laws* (2nd edn, Jordans 2002) 219. For Germany, see K Kuhne et al., ‘Germany’ in Nicholas, *French Law of Contract* 244–57.

¹⁹³ OY Truong, ‘Virtual inheritance: Assigning more virtual property rights’ (2009) 21(3) *Syracuse Science and Technology Law Reporter* 57, 57–86.

She proposes that the courts honour players' wishes to bequeath the value of their virtual assets and, if the players fail to do so, and the VW contract has a non-survivorship policy, then it will operate to bar claims by heirs.¹⁹⁴ Troung suggests that players should be able to transfer the non-monetary value of their virtual property to their immediate family members,¹⁹⁵ but they would only be able to transfer the whole account, not an individual item or monetary value, due to the conflicting interests of the VW provider. This solution is quite contradictory, as it argues for the transfer of virtual assets alone, not their monetary value, to abide by the contract and avoid conflict with the VW owner. However, at the same time, it violates universal contractual provisions' non-transferability of the entire account.

By contrast, the VWs usufruct is designed to minimise disruption to the VW ecology and conflict with the VW owner's wishes while recognising the VW player's earned right to transmit assets they have worked for in-game (see further below). First, the right will only be transmissible as a monetary claim if VA exchanges are legal on recognised auction sites. If no such auction sites exist or ToS do not permit them, then monetisation is impossible, and so is the right to compensation. Thus, the VW owner is not faced with an unwarranted financial burden as they will be compensated at a market rate for the virtual assets the owner created and thus not have to reach into their own pockets. The VW owner will have received during life either subscriptions or revenue via other means (e.g. adverts), so allowing the VW to retain monetisable VAs after a player's death could be seen as an unfair windfall.

Second, unlike Troung's proposal, the VW user's right respects typical VW contractual provisions forbidding the transfer of accounts, as family members do not get access to the deceased's account and cannot play the game instead of the deceased player. This minimises disruption to the game and its rules and loss of new revenue from new players. In order to further minimise this disruption, the monetary compensation described above would ideally need to be claimed within a particular time, for example six months or a year maximum. Possibly the right might also only be claimed by family members, as Troung suggests as well, to minimise disruption and burden to the VW further, although there seems no prevailing reason to make an exception to standard rules of freedom of testation (as defined in each legal system) for VAs. All the other assets would return to the VW, and to the players who are immersed in the VW and interested in making further use of these assets.

¹⁹⁴ Ibid. 80.

¹⁹⁵ Ibid.

This solution is a law reform proposal, and it would need to be enacted by relevant legislation in the individual jurisdictions, notwithstanding the practical difficulties of such a reform (e.g. in the US, provisions from the Fiduciary Access to Digital Assets Act to be enacted by the state laws; relevant legislation in the UK and other European countries). Consequently, it would entail changes in EULAs.¹⁹⁶

Advantages of VWs Usufruct

The first advantage of the VWs usufruct is that this model provides an acceptable compromise between the rights of the VW owner and those of the player. Practically, this is done in two ways: 1) recognising the respective contribution both of the capital to build the VW provided by the VW owner, as well as the money providers spend on maintenance and promotion of the world; and the labour as well as subscriptions and other monies provided by the players (e.g. money provided to purchase in-game assets); and 2) recognising the particular stake players have in ownership in VAs, even on death, which was explained above as relating to the constitutionalisation of VWs and the immersive environment in which players operate in-game.

The second significant advantage of this model is that it does not allow the heirs of a player to interfere with the operation of the VW after the player's death unless they pay to re-enter the game as new players with new accounts.

Third, the model does not interfere with standard provisions of VW contracts or EULAs forbidding the transfer of assets, accounts or passwords, except within recognised auction sites.

Fourth, this model does not provide for a general financial claim against the VW (either during the player's life or on death), which could be unduly burdensome if the VAs of the deceased could not be monetised.

Disadvantages of VWs Usufruct

First, the right suggested in principle operates only post-mortem. It does not require VWs to offer monetisation when a player leaves during life. Doing so might be seen as risky as potentially financially burdensome to existing VWs, thus discouraging growth and capital investment in future VWs. A compromise solution might be to apply the right during a lifetime to cases where the game is unjustifiably closed or destructively modified. This would not extend to justified improvements and developments of the VW, nor to bankruptcy

¹⁹⁶ It is worth noting, however, that it would be a matter for national legal systems to decide if the VW user's right could be excluded by contract. If this were allowed, it is likely the right would have little effect.

or similar where the VW owner had no choice but to close and assets are in any case likely to be seized by preferred creditors.

Second, the right might produce minimal and somewhat arbitrary benefits for players since many asset auction sites are illegal as they are unauthorised by the EULAs of the VWs. The issue then arises of whether VWs could be compelled by law to set up an authorised VA auction site, thus creating an additional burden on the provider. If the solution applies to the current state of play, the compromise would be reasonable from the VW owner's point of view, as they could effectively exclude it by not having an authorised auction site (and indeed, at present, there are few). Where there are recognised auction sites, service providers are already aware of this and approve of such a practice, so they can reasonably be expected to endorse the VW user's right. By corollary, players would have a reasonable expectation to realise and transfer the monetary value of their second layer VAs where authorised auction sites exist, so that value is something they would expect to transmit to their heirs as well.

Third, effective enforcement of the right may be difficult mainly when heirs and/or personal representatives are unaware of the VW in question or the game's environment in general. However, these difficulties have been overcome in relation to emails, as generally envisaged by the US Uniform Law Commission in the Uniform Fiduciary Access to Digital Assets Act (UFADAA).¹⁹⁷

Finally, as noted above, an issue could be potential contractual waivers if (as is likely) EULAs of VWs attempt to exclude the new VWs usufruct. Again, and as suggested in the RUFADAA, this can be overcome by prohibiting such provisions. While some practical aspects may need to be fine-tuned, the idea is presented here as one of the principles to produce a fairer balance of outcomes between players and VW developers.

It is worth noting that the solution differs from those suggested in other case studies in that there I do not suggest any default transmission of assets in the email and social network examples. As explained in the introductory part of this chapter, the main reason for this is that privacy is not as essential here since VAs do not typically embody or carry personally identifiable data. VW players typically disguise their identity under chosen pseudonyms or even assigned names, so assets they acquire, such as gold or virtual 'magic swords', do not reveal personal data. The monetised value of the assets, which

¹⁹⁷ National Conference of Commissioners on Uniform State Laws, Draft Fiduciary Access to Digital Assets Act, s 3 (July 2014) Drafting Committee Meeting <http://www.uniform-laws.org/shared/docs/Fiduciary%20Access%20to%20Digital%20Assets/2014am_ufadaa_draft.pdf> last accessed 1 August 2021.

will transfer, reveals still less about the deceased. Therefore, the interest in PMP asserted in this book does not prevent the suggested default but limited transmission of second layer assets.

Finally, it is worth noting that the solution here is in the form of a principle, without going into the technical details of succession law, economics or bankruptcy law. Instead, this chapter aims to provide guidance, based on the analysis of the previous literature on virtual property, taking into account the EULA provisions and special features of VWs. Chapter 7 will suggest general solutions applicable to all the case studies. It is also worth emphasising that the findings and proposal in this chapter do not apply to all types of video games, including downloaded games, games played 'locally', at the end user's side, or streamed games on Twitch. The reason is that these lack the essential features discussed herein, that is, physicality, constitutionalisation and citizenship. The model does, however, apply to any future games that resemble VWs and include a degree of virtuality, immersion, citizenship and other qualities discussed in this chapter.